

REPUBLIC OF TURKEY
ADANA ALPARSLAN TÜRKES SCIENCE AND TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
DEVELOPMENT OF AN AI-ENHANCED CSRR-BASED MICROWAVE SENSOR
FOR DIELECTRIC CHARACTERIZATION
CEM SONDUR
AKIN GEÇER
BACHELOR DEGREE PROJECT
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Assoc. Prof. Dr. Duygu Nazan Gençoğlu

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ABSTRACT

Development of an AI-Enhanced CSRR-Based Microwave Sensor for Dielectric Characterization

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This thesis presents the development of a simulation-based intelligent microwave sensing framework for dielectric characterization. The proposed system combines a two-port complementary split-ring-resonator-based microwave sensor, CST Studio Suite electromagnetic simulation, MATLAB automation, S21 feature extraction, and Random Forest regression. The central aim is not to claim direct chemical identification from a single scalar value, but to estimate the effective relative dielectric constant from the frequency-domain transmission response of a resonant microwave structure. This distinction is important because several materials may have close permittivity values, while real liquids may also exhibit loss, dispersion, temperature dependence, and sample-volume dependence.

The electromagnetic basis of the work is developed from Maxwell equations, constitutive relations, complex permittivity, transmission-line theory, scattering parameters, resonator quality factor, and cavity perturbation concepts. The sensing principle is that a dielectric sample placed in the high-field region of a resonant structure changes the effective stored electric energy and effective capacitance of the resonator. This perturbation shifts the resonance frequency and modifies the depth, bandwidth, slope, and area of the S21 resonance response. Therefore, the complete S21 curve contains richer information than the resonance frequency alone.

A MATLAB-CST automation workflow is used to generate a parametric dataset by sweeping the dielectric parameter ϵ_{degeri} from 1.0 to 10.0 with a step size of 0.1. For each simulated dielectric value, the two-port S21 response is exported as an ASCII file. MATLAB then reads the exported curves and extracts physically meaningful features such as resonance frequency, minimum transmission level, resonance depth, 3-dB bandwidth, loaded quality factor, frequency shift, resonance-area metric, mean response, standard deviation, and local slopes. These features form the input vector of a bagged regression-tree model. Cross-validation, RMSE, MAE, and R-squared are used to evaluate the predictive behavior of the model.

The resulting framework provides an end-to-end proof-of-concept pipeline for AI-enhanced microwave dielectric sensing. The completed simulation dataset contains 91 S21

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responses and the final regression model is evaluated with 5-fold cross-validation. The achieved validation results are RMSE = 0.1504 epsilon, MAE = 0.1045 epsilon, R-squared = 0.9967, and maximum absolute error = 0.5416 epsilon. These values demonstrate strong interpolation accuracy within the simulated epsilon interval, while the study remains a numerical calibration framework rather than a direct chemical identification system.

Keywords: CSRR, microwave sensor, dielectric characterization, S21, CST Studio Suite, MATLAB automation, Random Forest regression, resonator perturbation

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NOMENCLATURE

Table N.1. Nomenclature and abbreviations used throughout the thesis.

Symbol / Abbreviation	Definition
AI	Artificial Intelligence
BW	Bandwidth
CAD	Computer-Aided Design
COM	Component Object Model
CSRR	Complementary Split Ring Resonator
CST	CST Studio Suite
dB	Decibel
EM	Electromagnetic
f_0	Resonance frequency
MAE	Mean Absolute Error
MATLAB	Matrix Laboratory computing environment
Q	Quality factor
QL	Loaded quality factor
Q0	Unloaded quality factor
RF	Radio Frequency
RMSE	Root Mean Square Error
S11	Input reflection coefficient
S21	Forward transmission coefficient
VNA	Vector Network Analyzer
ϵ_r	Relative dielectric constant
ϵ_{eff}	Effective permittivity
$\tan \delta$	Loss tangent

1. INTRODUCTION

Microwave sensing is based on the interaction between electromagnetic fields and materials. When a material is placed in the sensitive region of a microwave resonator, the field distribution, stored electric energy, loss, and phase response of the structure are modified. These modifications are observed through measurable quantities such as resonance frequency, insertion loss, bandwidth, and quality factor. Because dielectric properties directly affect the electric displacement field, microwave resonators are attractive for non-destructive and label-free characterization of liquids and solids [1], [2].

The project presented in this thesis focuses on a two-port resonant microwave sensor whose response is evaluated through S21. In a two-port device, S21 represents the transmission from the input port to the output port. A resonance appears as a strong frequency-selective feature in the transmission curve. As the dielectric constant of the material under test changes, the resonant condition of the structure changes. The resulting S21 shift forms the physical basis of the sensing mechanism [1].

The engineering contribution of the work is the integration of electromagnetic simulation, automatic data generation, feature extraction, and machine learning into a single workflow. Instead of manually simulating a small number of material cases, MATLAB controls CST through automation, sweeps the dielectric parameter, exports S21 data, processes the resulting curves, and trains a regression model. This turns the microwave sensor into a computationally calibrated system. The final output of the model is an estimated effective relative dielectric constant rather than a direct chemical identity [4], [5], [11], [13].

A major design decision in this project is the use of S21 rather than S11 as the main sensing response. S11 is the reflection coefficient at the input port and is useful for one-port characterization. However, the sensor considered here is a two-port resonator, and its transmission behavior directly captures the frequency-selective energy transfer through the resonant structure. Therefore, the corrected data-generation workflow exports S2,1 from CST and all subsequent MATLAB processing is built around S21-based features [1], [4], [13].

1.1. Literature Review

The theoretical background of the present study is rooted in microwave network analysis and field theory. At RF and microwave frequencies, device dimensions can become comparable with wavelength, so distributed effects, port normalization, wave propagation, and scattering parameters must be treated explicitly rather than reduced to ordinary low-frequency circuit assumptions. Pozar provides the microwave-engineering basis for two-port S-parameters, resonators, loaded quality factor, and cavity perturbation, while Cheng

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provides the field-theoretic basis for Maxwell equations, material polarization, wave propagation, dielectric boundaries, and transmission-line behavior [1], [2].

Planar resonant sensors have become attractive because their localized electric fields can be concentrated around slots, gaps, rings, and complementary resonator geometries. A complementary split ring resonator can be interpreted as the slot-based counterpart of a split ring resonator, and its sensitivity is connected to the modification of effective capacitance and stored electric energy when a dielectric sample is placed near the high-field region. The use of complementary resonant structures is consistent with Babinet-related metamaterial interpretation and with previous microwave sensor studies on liquid and microfluidic dielectric characterization [6]-[8].

The literature also shows that a single resonance frequency is often not the only useful descriptor of a sensing response. Transmission magnitude, notch depth, bandwidth, quality factor, slope, and area-related quantities can contain additional information about coupling, loss, and resonance shape. For this reason, this thesis does not reduce the sensor response only to f_0 ; instead, the complete exported S21 curve is converted into a feature vector for regression. This design choice follows the general microwave measurement framework in which S-parameter curves represent calibrated port behavior over frequency [1], [4].

Full-wave electromagnetic simulation is necessary because the CSRR response depends on three-dimensional geometry, conductor boundaries, local dielectric loading, port definition, and mesh resolution. CST Studio Suite is used as the numerical electromagnetic environment, and its automation interface enables repeatable parameter sweeps and controlled S-parameter export. MATLAB is then used for data handling, feature extraction, model training, and statistical validation [4], [5], [13].

Machine learning has been used in several RF and microwave inverse-design or inverse-characterization problems because it can approximate nonlinear relationships between measurable responses and hidden physical parameters. Bishop provides the general pattern-recognition framework, Breiman introduced Random Forests as an ensemble of decorrelated decision trees, and recent microwave-oriented studies have used neural or tree-based approaches for antenna, component, and material-related inversion problems [3], [9]-[12].

The gap addressed in this thesis is therefore not the invention of a new electromagnetic law or a claim of direct chemical identification. The contribution is the construction of a controlled CST-MATLAB pipeline in which a CSRR-based two-port sensor is swept from $\epsilon_r = 1.0$ to 10.0, the correct S21 response is exported, physically interpretable S21 features are extracted, and a Random Forest regression model is validated as a numerical dielectric-calibration tool [3], [5], [11], [14].

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1.2. Motivation and Engineering Problem

Many practical liquids, polymers, substrates, and biological samples can be distinguished or monitored by their dielectric response at RF and microwave frequencies. The dielectric constant determines how strongly the material polarizes under an applied electric field, while the loss factor determines how much electromagnetic energy is dissipated as heat. In industrial and laboratory applications, it is often useful to monitor concentration, mixture ratio, purity, moisture, or material change without destructive chemical preparation [7], [8].

Conventional dielectric measurement can be performed with coaxial probes, waveguide fixtures, cavity resonators, or free-space methods. Each technique has trade-offs in terms of sample preparation, bandwidth, sensitivity, calibration complexity, and cost. Planar resonator sensors are attractive because they can be compact, low-cost, compatible with printed circuit fabrication, and sensitive to small material perturbations in localized high-field regions. The drawback is that their response can be nonlinear and geometry-dependent, which motivates computational calibration [1], [7], [8].

The engineering problem of this thesis can be stated as follows: given a two-port microwave resonator whose S21 response changes when the dielectric constant of a material changes, construct a reliable simulation-based workflow that maps S21 curve features to the effective dielectric constant. This requires accurate electromagnetic simulation, systematic parameter sweeping, correct export of transmission data, robust feature extraction, and a regression model that is evaluated with meaningful validation metrics [4], [5], [11].

1.3. Scope and Limitations of the Thesis

The scope of the present study is simulation-based dielectric characterization. The CST model provides the electromagnetic response, and MATLAB performs automation and learning. The model is not presented as a finished commercial instrument and it is not claimed to identify an unknown chemical material with absolute certainty. It estimates effective dielectric constant within the simulated range. This boundary is scientifically important because material identity depends not only on real permittivity but also on loss, frequency dispersion, sample volume, temperature, and measurement uncertainty [4], [5].

The simulated dielectric range is selected as $\epsilon_r = 1.0$ to 10.0 with 0.1 increments. This range is suitable for demonstrating the method on low-to-moderate permittivity materials and controlled dielectric loading. Materials with very high permittivity, highly lossy liquids, magnetic response, strong dispersion, or conductive behavior require additional modelling through complex permittivity and possibly frequency-dependent material models. The future-work chapter identifies these extensions [1], [2].

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Because physical fabrication and VNA measurement are not included at this stage, fabrication tolerances, connector repeatability, calibration-kit uncertainty, surface roughness variation, temperature drift, and liquid-holder errors are outside the current numerical dataset. The thesis therefore distinguishes between numerical model accuracy and physical instrument accuracy. This distinction prevents overclaiming and provides a clear route for experimental continuation [1].

1.4. Main Contributions

The first contribution is the corrected two-port S21-based data-generation pipeline. Earlier workflows can easily confuse S11 and S21 if the selected CST result-tree item is not checked carefully. In this thesis, the intended sensing observable is explicitly defined as S21, and the CST automation exports the S2,1 response for each dielectric value [1], [4], [13].

The second contribution is the use of multiple physically interpretable features rather than only the resonance frequency. The extracted feature vector includes f_0 , S_{min} , S_{max} , resonance depth, 3-dB bandwidth, loaded quality factor, frequency shift, resonance area, mean S21, standard deviation, and local slopes. These quantities describe not only where the resonance occurs but also how the resonance shape changes under dielectric loading [1], [7], [8].

The third contribution is the integration of Random Forest regression with validation metrics. The model is trained on S21-derived features and evaluated by cross-validation using RMSE, MAE, and R-squared. This provides a quantitative basis for discussing model performance in the results section after the simulations are completed [3], [5], [11].

2. MATERIALS AND METHODS

This chapter develops the theoretical and computational foundation of the thesis. The workflow begins with electromagnetic field theory because the sensor response is ultimately governed by Maxwell equations and material constitutive relations. The resonator is then interpreted through circuit and transmission-line concepts, because microwave engineering frequently expresses field behavior through port quantities such as impedance, reflection coefficient, transmission coefficient, and scattering parameters. After the theoretical foundation, the chapter describes CST modelling, MATLAB-CST automation, dataset generation, feature extraction, model training, and validation [4], [5], [13].

The central chain of causality used throughout the thesis is: dielectric loading changes the local electric energy distribution; the effective capacitance and effective permittivity of the resonant structure change; the resonance condition shifts; the S21 curve changes in frequency, depth, bandwidth, and slope; MATLAB extracts these changes as numerical

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features; and the regression model maps these features to effective relative permittivity [4], [5], [13].

2.1. Electromagnetic Field Basis

The electromagnetic operation of the sensor is governed by time-varying electric and magnetic fields. In macroscopic form, these fields are represented by electric field intensity E , magnetic field intensity H , electric flux density D , and magnetic flux density B . For linear isotropic media, D is related to E through permittivity and B is related to H through permeability. The electric field is the primary sensing field for dielectric characterization because dielectric polarization directly modifies D [1], [2].

For time-harmonic microwave problems, field quantities are commonly represented with phasors. A real time-domain field can be written as the real part of a complex phasor multiplied by $\exp(j \omega t)$. This representation transforms time derivatives into multiplication by $j \omega$ and makes frequency-domain simulation efficient. CST frequency-domain or time-domain solvers eventually produce frequency-domain port responses, which are the quantities exported to MATLAB [1], [2], [4].

The four Maxwell equations express the coupling between electric and magnetic fields, source currents, source charges, and material properties. In differential phasor form, the curl equations are the most directly relevant to wave propagation. Faraday law states that a time-varying magnetic field produces a rotational electric field, while Ampere-Maxwell law states that conduction and displacement currents produce a rotational magnetic field [2].

2.1.1. Maxwell Equations and Constitutive Relations

The following equations summarize the electromagnetic basis used in the thesis. The current density J may include conductive current, while the displacement term $j \omega D$ represents capacitive storage in the dielectric medium. In microwave dielectric sensing, this displacement current is especially important because the resonator field partly resides in the sample region [1], [4], [13].

In a linear isotropic nonmagnetic material, the relative permittivity ϵ_r determines the proportionality between D and E . If the material is lossless, ϵ_r can be treated as a real scalar. If the material is lossy, the permittivity is complex and the imaginary component represents energy dissipation. Real liquids often have frequency-dependent complex permittivity, but the present simulation stage uses a controlled real epsilon sweep unless complex material data are later introduced [1], [4], [13].

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$$\nabla \times \mathbf{E} = -j\omega\mathbf{B} \quad (1)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega\mathbf{D} \quad (2)$$

$$\mathbf{D} = \epsilon\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E} \quad (3)$$

$$\epsilon_r^* = \epsilon_r' - j\epsilon_r'' \quad (4)$$

$$\tan \delta = \frac{\epsilon_r''}{\epsilon_r'} \quad (5)$$

2.1.2. Wave Equation and Propagation Constant

Combining Maxwell curl equations in a homogeneous source-free region leads to the Helmholtz wave equation. The wave equation shows that the field distribution depends on angular frequency, permeability, permittivity, and conductivity. In a lossless dielectric, the phase constant increases with the square root of relative permittivity. Therefore, waves travel more slowly in materials with higher dielectric constant, and resonant structures effectively become electrically longer when loaded with a higher-permittivity sample [1], [2].

The propagation constant gamma combines attenuation and phase delay. Its real part alpha represents amplitude decay, and its imaginary part beta represents phase variation. For a low-loss dielectric, beta is approximately omega square-root mu epsilon, while alpha is related to dielectric loss. This is the connection between material properties and the measured microwave response [1], [2].

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$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0 \quad (6)$$

$$k = \omega \sqrt{\mu \epsilon} \quad (7)$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (8)$$

2.1.3. Complex Permittivity, Loss Tangent, and Dispersion

The relative permittivity of a material can be expressed as a complex quantity. The real component describes electric energy storage, while the imaginary component describes loss. The ratio of the imaginary part to the real part is the loss tangent. In dielectric sensing, a change in the real part mainly shifts the resonance frequency, while a change in the imaginary part generally broadens the resonance, reduces Q, and increases insertion loss [1], [2].

Frequency dispersion means that permittivity changes with frequency. Water, alcohols, biological tissues, and many polar liquids exhibit dispersion in microwave bands because dipolar relaxation mechanisms cannot follow arbitrarily high-frequency excitation. In the present project, the first calibration stage focuses on real-valued epsilon between 1 and 10. A more advanced stage can extend the model to estimate epsilon prime, epsilon double-prime, and tan delta simultaneously from full S21 curves [1], [4], [13].

2.2. Transmission-Line and Port Interpretation

Microwave circuits cannot always be interpreted with lumped circuit theory because voltage and current phase can vary over the physical dimensions of the structure. A planar resonator at gigahertz frequencies behaves as a distributed circuit: conductor traces, slots, gaps, and dielectric regions support electromagnetic waves and standing-wave patterns. Transmission-line theory provides the bridge between field solutions and measurable circuit quantities [1], [4], [13].

A transmission line is characterized by distributed resistance R, inductance L, conductance G, and capacitance C per unit length. These quantities lead to a propagation constant and characteristic impedance. When a line is terminated by a load different from its characteristic impedance, reflected waves occur. In a resonant sensor, the resonant

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discontinuity produces frequency-selective reflection and transmission effects that appear in the S-parameters [1], [4], [13].

Port-based interpretation is essential because CST exports the solution in terms of scattering parameters. These parameters are measured by a vector network analyzer in a physical experiment and are directly comparable to simulation. Thus, the same S21 feature extraction pipeline can later be applied to measured VNA data, provided that calibration and data formatting are handled correctly [4], [5], [13].

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (9)$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (10)$$

2.2.1. Scattering Parameters

Scattering parameters express the relationship between incident and reflected traveling waves at the ports of a microwave network. For a two-port network, S11 represents input reflection, S22 represents output reflection, S21 represents forward transmission, and S12 represents reverse transmission. In this thesis, S21 is the principal quantity because the device is treated as a two-port transmission-mode resonant sensor [1].

The use of S21 is also practical for resonance tracking. A resonator inserted or coupled between two ports produces a frequency-selective feature in the transmission curve. Depending on the coupling topology, the resonance may appear as a transmission dip or peak. The present MATLAB algorithm searches for a minimum in the S21 magnitude curve and extracts the frequency and shape parameters of that resonance. If the physical topology produces a peak instead of a dip, the feature extraction rule must be adjusted accordingly; this is why the thesis explicitly documents the assumed response shape [4], [5], [13].

In decibel form, the transmission magnitude is computed as $20 \log_{10}$ of the magnitude of S21. The dB scale is convenient because small and large transmission changes can be observed on the same plot. It also makes resonance depth and insertion-loss changes easier to compare across dielectric values [1], [4], [13].

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$$\mathbf{b} = \mathbf{S}\mathbf{a}, \quad S_{21} = \frac{b_2}{a_1} (a_2 = 0) \quad (11)$$

$$S_{21,dB} = 20\log_{10} |S_{21}| \quad (12)$$

2.3. Microwave Resonator Theory

A resonator stores electromagnetic energy by exchanging energy between electric and magnetic forms. In a lumped LC resonator, the electric energy is stored in the capacitor and the magnetic energy is stored in the inductor. In a planar microwave resonator, this storage is distributed across conductor edges, slots, gaps, substrate volume, and surrounding dielectric regions. The resonance frequency is determined by the effective inductance and effective capacitance of the structure [1].

When a dielectric sample is introduced into a region where the electric field is strong, the local electric energy storage increases. This is equivalent to increasing the effective capacitance of the resonator. Since the resonance frequency is inversely proportional to the square root of LC, increased effective capacitance generally shifts the resonance downward. This is the most important physical effect exploited in this thesis [1], [4], [13].

The quality factor Q measures how sharply the resonator stores energy relative to its energy loss. A high-Q resonator has a narrow bandwidth and high frequency selectivity, which can improve resolution but may also make the sensor more sensitive to fabrication tolerance and noise. A low-Q resonance is broader and may be easier to measure but less sensitive to small material changes. The extracted loaded Q in this work is obtained from f_0 divided by the 3-dB bandwidth of the S21 resonance feature [1], [4], [13].

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad (13)$$

$$Q_L = \frac{f_0}{BW_{3dB}} \quad (14)$$

2.3.1. Lumped Equivalent Model of a Resonant Sensor

Although the actual sensor is distributed, a lumped equivalent model helps explain the sensing mechanism. The resonant frequency of an equivalent LC circuit is determined by

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inductance and capacitance. In a CSRR-type structure, the slot and ring geometry contributes effective capacitance and inductance. The sample changes the effective capacitance because the electric field penetrates the region occupied by the sample [6]-[8].

Let C_0 represent the unloaded effective capacitance and ΔC represent the capacitance perturbation due to the sample. The loaded resonance frequency then decreases approximately with increasing ΔC . For small perturbations, a first-order expansion shows that the fractional frequency shift is approximately proportional to negative one-half of the fractional capacitance change. This explains why the frequency shift is a sensitive indicator of dielectric loading [1], [4], [13].

However, frequency shift alone does not capture the full response. Lossy samples can change the resonance depth and bandwidth without producing a unique frequency shift. Coupling strength can also alter insertion loss. Therefore, the thesis uses multiple features rather than a single f_0 value [1], [2].

2.3.2. Perturbation Theory for Dielectric Loading

Cavity and resonator perturbation theory provides a deeper mathematical explanation of dielectric sensing. If a small sample perturbs the permittivity distribution of a resonator, the resulting fractional shift in resonant frequency is related to the electric energy stored in the perturbed volume relative to the total stored energy. The stronger the electric field inside the sample region, the larger the frequency shift [1], [2].

The perturbation relation also explains the importance of sample placement. A dielectric sample placed at an electric-field maximum produces a strong response, whereas the same sample placed at an electric-field minimum produces weak sensitivity. Therefore, the sensor geometry should be designed so that the material-under-test region overlaps with intense electric fields. In a CSRR sensor, high electric fields commonly occur around the slot gaps and ring discontinuities [1], [4], [13].

The perturbation formula assumes that the perturbation is small and that the field distribution is not radically changed. For larger samples or high-permittivity materials, full-wave simulation is required because the field distribution and coupling condition may change nonlinearly. This justifies the use of CST for dataset generation rather than relying only on closed-form equations [1], [4], [13].

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$$\frac{\Delta f}{f_0} \approx -\frac{1}{2} \frac{\Delta C}{C_0} \quad (15)$$

$$\frac{\Delta f}{f_0} \approx -\frac{1}{2} \frac{\int_{V_s} \Delta \epsilon |\mathbf{E}|^2 dV}{\int_V (\epsilon |\mathbf{E}|^2 + \mu |\mathbf{H}|^2) dV} \quad (16)$$

2.3.3. Loaded and Unloaded Quality Factor

The unloaded quality factor Q_0 describes the intrinsic energy storage and loss of the resonator without external loading. In practice, any resonator connected to ports is loaded by measurement circuitry, feed lines, coupling gaps, and termination impedances. The measured quality factor is therefore the loaded quality factor Q_L . In the present thesis, Q_L is extracted from the S_{21} curve because that is the available two-port response [1], [4], [13].

The relationship between unloaded Q , external Q , and loaded Q depends on coupling. Strong coupling lowers the loaded Q and may increase transmitted power at resonance. Weak coupling preserves high Q but may reduce signal level. For sensing, the optimum is not always the highest Q ; the useful criterion is repeatable and monotonic feature change with dielectric loading. A resonance that is too shallow may be difficult to track, while a resonance that is too narrow may require very fine frequency resolution [1], [2].

In the MATLAB workflow, the 3-dB bandwidth around the resonance minimum is estimated from the S_{21} curve. The loaded Q is then computed as f_0 divided by bandwidth. This is a practical feature for machine learning because it captures resonance broadening and narrowing as the dielectric value changes [4], [5], [13].

2.4. Complementary Split Ring Resonator Sensor Principle

A complementary split ring resonator is the negative or slot-based counterpart of a split ring resonator. In planar microwave sensors, CSRR structures are often etched in the ground plane or conductor layer and excited by a nearby transmission line. The slot geometry creates localized electric fields and a resonant stopband or transmission feature. The resonance is sensitive to the dielectric properties of material placed close to the slot [6]-[8].

The complementary nature of the CSRR means that the electromagnetic response is strongly associated with electric-field concentration across the etched gaps. This makes CSRR sensors suitable for dielectric loading because the dielectric sample directly modifies the capacitance associated with the slot. In a two-port implementation, this

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perturbation is observed through the S21 transmission response between the input and output ports [1], [4], [13].

The CST model used in this project contains a resonant structure with a dielectric region controlled by a parameter named `eps_degeri`. During automation, MATLAB changes `eps_degeri`, rebuilds the model, runs the solver, selects S2,1 from the result tree, and exports the corresponding S21 curve. This controlled parameter sweep converts the electromagnetic model into a numerical calibration dataset [4], [5], [13].

2.5. Numerical Electromagnetic Simulation in CST

Full-wave simulation solves Maxwell equations for the actual three-dimensional structure, including conductor geometry, substrate, ports, dielectric loading region, and boundary conditions. This is necessary because CSRR sensors generally do not have a simple closed-form solution. The fields are strongly affected by slot dimensions, coupling distance, substrate permittivity, finite metal thickness, and port definition [4].

CST Studio Suite provides the electromagnetic solver environment for this thesis. The user-defined material parameter `eps_degeri` controls the relative permittivity of the dielectric sample region. By changing this parameter and rebuilding the model, the same geometry can be evaluated for many dielectric values. This is a controlled simulation experiment: the geometry is kept fixed while the material parameter is varied systematically [1], [4], [13].

The simulation output is the frequency-domain S21 curve. The exported ASCII file contains frequency and transmission data. MATLAB reads this file, normalizes frequency units when necessary, converts linear magnitude to dB when necessary, and extracts resonance features. This separation of simulation and post-processing makes the workflow reproducible and easier to debug [4], [5], [13].

2.5.1. Boundary Conditions, Ports, and Meshing Considerations

The accuracy of a microwave simulation depends on port definition, boundary conditions, and mesh resolution. Ports must excite the intended quasi-TEM or microwave mode without introducing artificial mismatch. Boundaries must be placed far enough or selected appropriately so that unwanted reflections do not distort the resonant response. Mesh density must be sufficient near narrow slots, gaps, and high-field regions because these locations dominate the capacitance and sensitivity [1], [4], [13].

In resonator sensing, small geometric details can shift the resonance frequency. Therefore, the mesh should resolve CSRR slots and sample boundaries carefully. If the mesh is too coarse, the computed capacitance and field concentration may be inaccurate. If the mesh is too fine, simulation time becomes excessive. This thesis uses CST as the

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primary solver but documents the need for mesh-awareness as a limitation and future optimization topic [1], [4], [13].

A practical simulation workflow also requires checking whether the resonance selected by MATLAB is the physically intended mode. When dielectric loading changes, multiple resonances may appear or the global minimum may jump to a different mode. To reduce this risk, the final MATLAB code includes a resonance-tracking window around the previous f_0 . This is an engineering safeguard against mode misidentification [4], [5], [13].

2.6. MATLAB-CST Automation Workflow

The automation workflow is one of the central engineering components of the thesis. Instead of manually changing the dielectric constant and exporting data for each simulation, MATLAB controls CST through the ActiveX/COM interface. The script opens the CST file, stores a new value for `eps_degeri`, rebuilds the model, starts the solver, selects the S2,1 result, and exports the S21 curve to a text file [13].

The epsilon sweep is defined as 1.0:0.1:10.0, resulting in 91 simulated dielectric values. Each exported file is named according to the dielectric value, for example `eps_1.00.txt`, `eps_1.10.txt`, and `eps_10.00.txt`. This naming convention allows MATLAB to recover the target epsilon value directly from the file name during model training [4], [5], [13].

The final automation code also includes checks that were missing in the early prototype. It verifies that the exported file exists and is not empty. It uses the S2,1 tree path rather than S1,1. It removes hard-coded user paths and uses the project folder `C:\Users\CEM\Desktop\Bitirme2` as the default, with interactive folder selection as a fallback. These improvements make the workflow more robust and easier to reproduce [4], [5], [13].

2.6.1. CST Dataset Generation Code

The following MATLAB excerpt shows the essential structure of the corrected CST automation stage. The code is written without explanatory comment lines inside the block and uses English variable names for thesis presentation. The full executable file is maintained separately as `CST_S21_VeriSeti_Olustur_Nihai.m` [4], [13].

Code 2.1. Core MATLAB-CST automation excerpt for S21 dataset generation.

```
clear; clc; close all
projectRoot = 'C:\Users\CEM\Desktop\Bitirme2';
cstFile = fullfile(projectRoot, 'dielektrik ile rezonatör.cst');
exportFolder = fullfile(projectRoot, 'S21_Exports');
epsilonValues = 1.0:0.1:10.0;
parameterName = 'eps_degeri';
sParameterPath = '1D Results\S-Parameters\S2,1';
if ~exist(exportFolder, 'dir')
    mkdir(exportFolder)
end
cst = actxserver('CSTStudio.application');
invoke(cst, 'OpenFile', cstFile);
```

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```

model = invoke(cst, 'Active3D');
for index = 1:numel(epsilonValues)
    epsilonValue = epsilonValues(index);
    invoke(model, 'StoreDoubleParameter', parameterName, epsilonValue);
    invoke(model, 'Rebuild');
    solver = invoke(model, 'Solver');
    invoke(solver, 'Start');
    pause(1.0)
    invoke(model, 'SelectTreeItem', sParameterPath);
    exportPath = fullfile(exportFolder, sprintf('eps_%.2f.txt', epsilonValue));
    asciiExport = invoke(model, 'ASCIIExport');
    invoke(asciiExport, 'Reset');
    invoke(asciiExport, 'FileName', exportPath);
    invoke(asciiExport, 'Execute');
end
invoke(model, 'Save');
invoke(cst, 'Quit');
delete(cst);

```

2.7. S21 Data Reading and Feature Extraction

After simulation, each S21 text file is processed in MATLAB. The first task is to read frequency and transmission data robustly. CST exports may contain frequency in GHz, MHz, or Hz depending on settings, and S-parameter data may appear as dB magnitude, linear magnitude, or complex real-imaginary values. The final MATLAB code includes logic to normalize these possibilities before feature extraction [4], [5], [13].

Feature extraction begins by locating the resonance minimum in the S21 curve. To reduce sensitivity to numerical noise, the curve may be smoothed using a moving median operation. The algorithm then computes the resonance frequency f_0 , minimum S21 value, maximum S21 value, depth, 3-dB bandwidth, loaded Q, area below the maximum response, mean S21, standard deviation, and local slopes on both sides of the resonance. These features convert the whole curve into a compact numerical signature [4], [5], [13].

The use of multiple features is mathematically justified because the S21 curve is not completely described by f_0 . Two different dielectric cases could theoretically produce similar f_0 values but different loss or coupling characteristics. Bandwidth and depth help capture these differences. Local slopes and area metrics describe resonance shape. Therefore, the feature vector is more informative than a single resonance-frequency scalar [4], [5], [13].

2.7.1. Feature Definitions

The resonance frequency f_0 is the frequency at which the selected S21 dip reaches its minimum. S_{min} is the value of S21 at f_0 . S_{max} is the maximum value in the searched frequency band. The depth is S_{max} minus S_{min} and represents how strong the resonant rejection or transmission contrast is. The 3-dB bandwidth is measured between the left and right frequencies where the curve rises 3 dB above the minimum. Loaded Q is f_0 divided by this bandwidth [1], [4], [13].

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The frequency shift Δf is computed relative to the reference resonance, typically the lowest epsilon case or air-like case. The area below maximum is the integral of S_{max} minus S_{21} over frequency, which captures both resonance width and depth. Mean and standard deviation describe the overall transmission level and variability in the frequency band. Local slopes describe the left and right sharpness of the resonance [1], [4], [13].

These features are physically interpretable. f_0 and Δf relate to effective permittivity and capacitance. S_{min} and depth relate to coupling and loss. Bandwidth and Q relate to stored energy and dissipation. Area and slopes describe the distributed shape of the resonance. The feature vector therefore connects microwave physics with machine learning [4], [5], [13].

$$\Delta f = f_0(\epsilon_r) - f_0(\epsilon_{ref}) \quad (17)$$

$$\mathbf{x} = [f_0, S_{min}, S_{max}, D, BW, Q_L, \Delta f, A, \mu_S, \sigma_S, m_L, m_R]^T \quad (18)$$

Table 2.1. Extracted S21 features and physical interpretation.

Feature	Physical meaning
F0_GHz	Resonance position
Smin_dB	Minimum transmission at resonance
Depth_dB	Resonance contrast
Bandwidth3dB_GHz	Width of resonance around 3-dB level
LoadedQ	Frequency selectivity
DeltaF_GHz	Shift relative to reference
AreaBelowMax	Combined width-depth metric
LeftSlope / RightSlope	Sharpness of resonance sides

2.7.2. Feature Extraction Code

The following code excerpt summarizes the feature extraction logic used in MATLAB. It avoids comment lines and uses English variable names in the thesis version. The algorithm can process either simulated S21 files or future measured VNA files if the file format is compatible [4], [5], [13].

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Code 2.2. Core MATLAB feature extraction excerpt.

```
function featureTable = extractS21FeaturesFromFolder(exportFolder)
files = dir(fullfile(exportFolder, 'eps_*.txt'));
rows = {};
referenceF0 = [];
previousF0 = [];
for fileIndex = 1:numel(files)
    fileName = files(fileIndex).name;
    filePath = fullfile(files(fileIndex).folder, fileName);
    epsilonValue = sscanf(fileName, 'eps_%.txt');
    data = readmatrix(filePath, 'FileType', 'text');
    frequencyGHz = normalizeFrequency(data(:,1));
    s21dB = normalizeS21(data);
    features = calculateResonanceFeatures(frequencyGHz, s21dB, previousF0);
    previousF0 = features.F0_GHz;
    if isempty(referenceF0)
        referenceF0 = features.F0_GHz;
    end
    rows(end+1,:) = {string(fileName), epsilonValue, features.F0_GHz, features.Smin_dB,
features.Smax_dB, features.Depth_dB, features.Bandwidth3dB_GHz, features.LoadedQ,
features.F0_GHz - referenceF0, features.AreaBelowMax, features.MeanS21_dB, features.StdS21_dB,
features.LeftSlope_dBperGHz, features.RightSlope_dBperGHz};
end
featureTable = cell2table(rows, 'VariableNames', {'FileName','Epsilon','F0_GHz','Smin_dB','Smax_
dB','Depth_dB','Bandwidth3dB_GHz','LoadedQ','DeltaF_GHz','AreaBelowMax','MeanS21_dB','StdS21_dB'
,'LeftSlope_dBperGHz','RightSlope_dBperGHz'});
end
```

2.8. Machine Learning Model

The regression problem is defined as estimating ϵ_r from an input feature vector x extracted from the S21 response. The training set consists of pairs $(x_i, \epsilon_{r,i})$, where $\epsilon_{r,i}$ is known from the CST parameter sweep. The objective is to learn a function that generalizes from simulated feature vectors to unseen feature vectors within the same dielectric range [3], [11].

A bagged regression-tree ensemble is used in MATLAB through fitensemble with Method set to Bag. This model is closely related to Random Forest regression. Each tree partitions the feature space into regions and predicts a constant or averaged target value in each terminal region. Bagging trains many trees on bootstrap samples and averages their predictions. Averaging reduces variance and improves stability compared with a single decision tree [3], [5], [11], [14].

Tree ensembles are suitable for this problem because the relation between S21 features and epsilon may be nonlinear. A simple linear model may not capture resonance-shape changes across the full dielectric range. A neural network could also be used, but for a moderate simulation dataset, a bagged tree ensemble is easier to train, less sensitive to scaling, and more interpretable through feature importance analysis [3], [5], [11], [14].

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$$D = \{(\mathbf{x}_i, \varepsilon_i)\}_{i=1}^N \quad (19)$$

2.8.1. Regression Trees and Bagging Mathematics

A regression tree recursively splits the feature space by selecting thresholds that reduce prediction error. At a given node, the algorithm evaluates candidate splits and chooses the split that minimizes the sum of squared errors in the child nodes. The terminal-node prediction is commonly the average target value of training samples that fall into that region. This creates a piecewise approximation of the unknown mapping from S21 features to epsilon [3], [11].

A single tree can have high variance because small changes in the training set can change the selected split thresholds. Bagging addresses this by drawing bootstrap samples from the training set and training one tree per sample. The ensemble prediction is the average of the individual tree predictions. This averaging process keeps nonlinear modelling ability while reducing sensitivity to data perturbations [3], [5], [11], [14].

In this thesis, the ensemble is not used as a black box only. Its input features are engineered from microwave theory, and its output is a physical parameter. This makes the learning problem more constrained and meaningful than feeding arbitrary raw data into a model. The model learns a calibrated mapping over a defined dielectric interval [3], [5], [11], [14].

$$J(s) = \sum_{i \in R_L(s)} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R(s)} (y_i - \bar{y}_R)^2 \quad (20)$$

$$\hat{\varepsilon}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}) \quad (21)$$

2.8.2. Validation Metrics

Model evaluation is performed using cross-validation. If the dataset is small, leave-one-out cross-validation is used because it allows each sample to serve once as a test case. If the dataset has enough samples, k-fold cross-validation is used. The final 1.0 to 10.0 sweep with 0.1 spacing is expected to provide 91 samples, so k-fold validation is appropriate [3], [5].

RMSE penalizes larger errors more strongly because the errors are squared before averaging. MAE gives the average absolute error and is easier to interpret directly in

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epsilon units. R-squared measures how much of the target variance is explained by the model relative to a constant mean predictor. These metrics should be reported together because each emphasizes a different aspect of performance [3], [5], [11], [14].

The model should not be judged only by a high R-squared value. The S21 curves and f0-epsilon trend must also be physically inspected. If the resonance tracker selects the wrong mode, the model may learn an artificial relationship. Therefore, visual plots are part of the validation workflow: S21 families, f0-epsilon calibration, predicted-versus-true epsilon, and error distribution are all required for the final results chapter [3], [5], [11], [14].

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\epsilon_i - \hat{\epsilon}_i)^2} \quad (22)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |\epsilon_i - \hat{\epsilon}_i| \quad (23)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\epsilon_i - \hat{\epsilon}_i)^2}{\sum_{i=1}^N (\epsilon_i - \bar{\epsilon})^2} \quad (24)$$

2.8.3. Model Training Code

The following code excerpt shows the core model training and validation logic. It is intentionally presented without MATLAB comment lines and with English variable names. The full file is maintained separately as AkilliSensor_ModelEgitimi_Nihai.m [5], [11].

Code 2.3. Core MATLAB model training excerpt.

```
featureNames = {'F0_GHz','Smin_dB','Smax_dB','Depth_dB','Bandwidth3dB_GHz','LoadedQ','DeltaF_GHz',
'AreaBelowMax','MeanS21_dB','StdS21_dB','LeftSlope_dBperGHz','RightSlope_dBperGHz'};
inputTable = featureTable(:, featureNames);
targetVector = featureTable.Epsilon;
treeTemplate = templateTree('MinLeafSize', 2, 'Surrogate', 'on');
trainedModel = fitrensemble(inputTable, targetVector, 'Method', 'Bag', 'NumLearningCycles', 300,
'Learners', treeTemplate);
if height(featureTable) < 30
    validationModel = crossval(trainedModel, 'Leaveout', 'on');
    validationType = 'Leave-One-Out Cross Validation';
else
    validationModel = crossval(trainedModel, 'Kfold', 5);
    validationType = '5-Fold Cross Validation';
end
predictedVector = kfoldPredict(validationModel);
errorVector = predictedVector - targetVector;
rmseValue = sqrt(mean(errorVector.^2));
maeValue = mean(abs(errorVector));
r2Value = 1 - sum((targetVector - predictedVector).^2) / sum((targetVector -
mean(targetVector)).^2);
save('AkilliSensor_Model_Nihai.mat', 'trainedModel', 'featureNames', 'rmseValue', 'maeValue',
```

```
'r2Value', 'validationType', 'featureTable');
```

2.9. Reproducibility and File Organization

The project is organized around the root folder C:\Users\CEM\Desktop\Bitirme2. The CST model file is named dielektrik ile rezonatör.cst. The S21 export folder is S21_Exports. Model outputs are written to Nihai_Model_Ciktilari. This organization separates raw simulation data, extracted features, trained model files, validation graphics, and interface files [1], [4], [13].

Reproducibility requires that each exported file can be traced back to its dielectric value. The eps_*.txt naming convention provides this traceability. The feature table stores file name, epsilon value, and extracted features in a CSV file. The model MAT file stores the trained ensemble, feature names, metrics, and configuration. This makes it possible to revisit the model later without rerunning every CST simulation [3], [5], [11], [14].

$$\epsilon_r \in \{1.0, 1.1, 1.2, \dots, 10.0\}, \quad N = 91 \quad (25)$$

Table 2.2. Reproducible project configuration.

Item	Value
Project root	C:\Users\CEM\Desktop\Bitirme2
CST file	dielektrik ile rezonatör.cst
Export folder	S21_Exports
Epsilon range	1.0 to 10.0
Step size	0.1
Number of simulations	91
Primary S-parameter	S21 / S2,1
Model output	Effective relative dielectric constant

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2.10. Extended Mathematical Interpretation of Sensitivity

Sensitivity can be interpreted as the derivative of the measurable feature with respect to dielectric constant. For a frequency-shift sensor, the primary sensitivity is $S_f = d f_0 / d \epsilon_r$. In practice this derivative is estimated from the discrete calibration curve. A large absolute value of S_f means that a small change in permittivity produces a measurable frequency movement. However, sensitivity alone is not sufficient; the resonance must also be stable, repeatable, and separable from numerical or measurement noise [1], [7], [8].

The frequency resolution of the simulated or measured S21 data limits the smallest detectable frequency shift. If the frequency step is too coarse, the extracted f_0 may jump between grid points, creating quantization error. Interpolation around the resonance minimum can reduce this error, but interpolation should not hide incorrect mode tracking. A robust final implementation can use polynomial fitting around the dip or Lorentzian fitting to estimate f_0 continuously [1], [4], [13].

The quality factor affects sensitivity interpretation. A high-Q resonance provides a steep slope near resonance, making small frequency shifts easier to detect. However, if Q is too high, the sweep must have sufficiently fine resolution to sample the dip accurately. If the resonance is broad, the dip is easier to sample but the same frequency shift may be less visually distinct. Therefore, sensor performance depends on the combined behavior of sensitivity, Q, depth, and noise level [1], [2].

A shape-based feature vector also supports model robustness. Suppose two curves have nearly identical f_0 but different bandwidths. A scalar f_0 model would map them to the same epsilon value, while a multi-feature model can use bandwidth, depth, and area to separate them. This is especially important when future experimental data include loss and temperature effects [4], [5], [13].

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$$S_f = \frac{df_0}{d\epsilon_r} \quad (26)$$

$$S_{norm} = \frac{1}{f_0} \left| \frac{df_0}{d\epsilon_r} \right| \quad (27)$$

$$\Delta f_{min} \geq \max(\Delta f_{grid}, \Delta f_{noise}, \Delta f_{fit}) \quad (28)$$

2.11. Error Sources and Uncertainty Propagation

Even in simulation, several sources of uncertainty can affect the final model. Mesh discretization changes the numerical approximation of the electromagnetic fields. Port definition affects the computed S-parameters. Solver convergence affects curve smoothness. Frequency sampling affects the accuracy of the extracted resonance. In physical measurements, additional uncertainties arise from VNA calibration, connector repeatability, fabrication tolerance, sample placement, temperature, and liquid evaporation [1], [4], [13].

Uncertainty propagation can be considered by treating the extracted features as noisy variables. If epsilon is predicted by a function $g(x)$, then small feature perturbations produce output perturbations approximately described by the gradient of g with respect to x . For nonlinear tree ensembles, the gradient is not smooth everywhere, but local sensitivity can still be evaluated numerically by perturbing each feature and observing the change in predicted epsilon [3], [5], [11], [14].

A future experimental version should repeat measurements for each reference liquid and report mean and standard deviation of predicted epsilon. Repeated measurement is necessary because a single S21 curve cannot capture repeatability. The present simulation model establishes the computational structure required for this later uncertainty analysis [1], [4], [13].

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$$\sigma_{\varepsilon}^2 \approx \nabla g(\mathbf{x})^T \Sigma_x \nabla g(\mathbf{x}) \quad (29)$$

$$\delta_j = \frac{g(\mathbf{x} + h\mathbf{e}_j) - g(\mathbf{x} - h\mathbf{e}_j)}{2h} \quad (30)$$

2.12. Why Full-Wave Simulation is Necessary

The simplified LC and perturbation equations explain the direction and approximate magnitude of the sensing effect, but they cannot fully predict the response of the actual geometry. CSRR structures include distributed fields, finite conductors, dielectric substrates, port fringing, coupling gaps, and radiation or boundary effects. These effects interact in a way that generally requires full-wave electromagnetic simulation [1], [4], [13].

Full-wave simulation is also necessary because the resonant field is not uniformly distributed. The sample may occupy only part of the sensitive field region. Therefore, the effective permittivity experienced by the resonator is not simply the sample permittivity; it is weighted by the field distribution. This is why the model output is described as effective dielectric constant estimation [1], [4], [13].

A second reason for simulation is that the S21 response contains coupling information. Coupling is determined by feedline position, geometry, substrate, and resonator loading. Analytical perturbation formulas usually assume an ideal resonant mode and do not directly include practical port coupling and insertion loss. CST includes these effects in the computed S-parameters [1], [4], [13].

2.13. Interpretation of the Learning Problem as Calibration

The machine-learning model should be understood as a calibration function. It does not discover permittivity from first principles alone; rather, it learns the inverse mapping created by the simulated sensor. The forward mapping is electromagnetic: epsilon produces an S21 curve. The inverse mapping is computational: S21 features predict epsilon. Calibration is necessary because the inverse mapping may be nonlinear and geometry-specific [1], [4], [13].

This interpretation also clarifies the limits of extrapolation. A model trained on epsilon values between 1 and 10 should not be trusted outside this interval without additional simulations. Tree-based models are especially weak at extrapolation because their predictions are based on regions formed within the training feature space. Therefore, the model deployment interface should warn users when the input resonance lies outside the calibration range [1], [4], [13].

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The calibration view also supports future transfer to measurement. Once a physical sensor is fabricated, measured reference liquids can be added to or used to correct the CST-derived dataset. A hybrid dataset containing both simulated and measured curves would allow the model to learn systematic differences between simulation and hardware [1], [4], [13].

2.14. Expanded Mathematical Foundation

The following additional subsections deepen the theoretical foundation of the project. They are inserted before the results chapter because the final numerical results must be interpreted through electromagnetic field theory, resonator perturbation theory, S-parameter analysis, and supervised regression mathematics. The purpose of this expansion is to make the thesis defensible not only as a MATLAB-CST implementation, but also as a complete engineering study with a traceable mathematical basis [4], [5], [13].

2.15. Detailed Field-Theoretic Development of the Sensor Response

The sensing mechanism can be developed more rigorously by starting from the time-harmonic form of Maxwell equations. In a source-free region of the simulated resonator, the electric and magnetic fields are represented by phasors. The physical time-domain field is the real part of the phasor multiplied by the exponential time factor. This representation is used because CST and S-parameter measurements operate in the frequency domain. The material under test enters the field equations through the constitutive relation between electric flux density and electric field [1], [4], [13].

When the material permittivity changes, the field solution of the entire boundary-value problem changes. The effect is not limited to the small region occupied by the sample. The perturbation changes local displacement current, modifies charge accumulation near the resonator slot or gap, and consequently changes the equivalent capacitance of the resonant structure. Because the resonant frequency is determined by the balance between stored electric and magnetic energy, a change in the electric energy term appears as a measurable frequency shift [1], [4], [13].

The full-wave problem solved by CST may be interpreted as an eigenvalue or driven-frequency problem. In an eigenvalue view, the resonator supports natural modes whose resonant frequencies depend on geometry and materials. In a driven two-port view, the resonator is excited through ports and the response is observed as a transfer function. The thesis uses the second view because S21 is directly exported from a port-based CST simulation. The eigenvalue interpretation is still useful for explaining why resonance shifts with dielectric loading [1], [4], [13].

At resonance, the electromagnetic structure stores energy alternately in electric and magnetic forms. A dielectric sample placed in a high electric-field region increases electric

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energy storage because the electric flux density becomes larger for the same electric field. This is why the high-field region of a CSRR or resonator gap is a sensitive sampling region. If the sample is placed in a weak-field region, the integral contribution of the sample to the total stored energy is small and the sensor sensitivity decreases [6]-[8].

The mathematical reason for using field-weighted integrals is that microwave sensing is not determined by material volume alone. Two samples with the same volume and the same dielectric constant can produce different shifts if they occupy different field regions. Therefore, the measured response is an effective property of the sample, its location, the resonator mode, and the surrounding geometry. This also explains why the trained model must be calibrated for the specific CST geometry used in this thesis [1], [4], [13].

$$\mathbf{E}(\mathbf{r}, t) = \Re\{\mathbf{E}(\mathbf{r})e^{j\omega t}\} \quad (31)$$

$$\mathbf{D}(\mathbf{r}) = \varepsilon(\mathbf{r})\mathbf{E}(\mathbf{r}) \quad (32)$$

$$W_e = \frac{1}{4} \int_V \varepsilon(\mathbf{r}) |\mathbf{E}(\mathbf{r})|^2 dV \quad (33)$$

$$W_m = \frac{1}{4} \int_V \mu(\mathbf{r}) |\mathbf{H}(\mathbf{r})|^2 dV \quad (34)$$

2.16. Dielectric Polarization, Complex Permittivity, and Loss Mechanisms

A dielectric material responds to an applied electric field by polarization. Polarization represents the displacement or reorientation of bound charges inside the material. In a simple low-loss material, polarization follows the applied field almost in phase. In real liquids and solids, the response may lag behind the field because molecular rotation, ionic conduction, and interfacial polarization require finite time. This lag is represented mathematically by complex permittivity [1], [2].

Complex permittivity has a real component and an imaginary component. The real component determines how much electric energy can be stored in the material. The imaginary component represents dielectric loss and contributes to attenuation. In a microwave resonator, the real part is mainly associated with resonance-frequency shift, while the imaginary part can broaden the resonance, reduce the depth, and lower the quality

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factor. Therefore, a future extension of this work should estimate both real and imaginary permittivity rather than only an effective real epsilon value [1], [2].

The loss tangent is the ratio of imaginary permittivity to real permittivity. A small loss tangent indicates that the material stores energy efficiently with little dissipation. A large loss tangent indicates that the material absorbs microwave energy strongly. In S21 measurements, loss can appear as increased insertion loss and broader resonance. If a model uses only f_0 , it may ignore loss-related information. The multi-feature approach in this thesis is therefore more physically meaningful because it includes depth, bandwidth, and area features [1], [4], [13].

Dispersion is another important issue. Dielectric constant may vary with frequency, especially in polar liquids. The CST sweep used in the current stage treats ϵ_{degeri} as a controlled scalar parameter. This is adequate for developing and validating the simulation-to-learning workflow, but it is not equivalent to a complete broadband complex-permittivity model. In experimental work, the permittivity should be associated with the operating frequency range and temperature [1], [4], [13].

For the purpose of this thesis, the target variable is the effective relative dielectric constant. The word effective is important. It means the model predicts the permittivity value represented by the CST parameter and sensed by the resonator under the defined geometry. It does not claim that the system independently separates all physical mechanisms in a real unknown liquid [1], [4], [13].

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$$\varepsilon^*(\omega) = \varepsilon'(\omega) - j\varepsilon''(\omega) \quad (35)$$

$$\varepsilon' = \varepsilon_0 \varepsilon'_r, \quad \varepsilon'' = \varepsilon_0 \varepsilon''_r \quad (36)$$

$$\tan \delta = \frac{\varepsilon''}{\varepsilon'} \quad (37)$$

$$P_d = \frac{1}{2} \omega \varepsilon'' |\mathbf{E}|^2 \quad (38)$$

2.17. Equivalent Circuit Interpretation of CSRR-Based Sensing

Although the actual sensor is a distributed electromagnetic structure, an equivalent circuit helps explain the trend of the response. A resonator can be represented near resonance by an inductive element, a capacitive element, and a loss element. In a complementary split-ring resonator, the slots and gaps concentrate electric field and behave like capacitive regions, while the current paths and surrounding conductor geometry contribute inductance. The substrate and sample modify the effective capacitance [6]-[8].

The equivalent capacitance is not a single physical capacitor. It is a compact representation of distributed electric energy stored around the resonator. When the dielectric constant of the sample increases, the electric energy stored in the same field region increases. In the LC analogy, this corresponds to an increase in effective capacitance. Because resonant frequency is inversely proportional to the square root of capacitance, the frequency generally shifts downward for stronger dielectric loading [1], [4], [13].

The resistance in the equivalent circuit represents losses. These include conductor loss, dielectric loss, radiation loss, and external loading through the feed lines. In a two-port configuration, the measured Q is the loaded Q because the resonator is coupled to ports. The unloaded Q would describe the resonator itself without external loading, but direct measurement of unloaded Q is not available from the exported S21 curve alone unless additional coupling analysis is performed [1], [4], [13].

The equivalent circuit is valid only near the selected resonance. Away from resonance, higher-order modes, parasitic capacitances, distributed phase delay, and radiation effects

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can make a simple LC model inaccurate. This is one reason why the thesis combines theoretical explanation with full-wave simulation. The LC model provides intuition; CST provides the numerical frequency response; MATLAB extracts the measurable descriptors [4], [5], [13].

The model features correspond naturally to equivalent-circuit quantities. The resonance frequency is linked to the LC product. The bandwidth and loaded Q are linked to energy loss and coupling. The resonance depth is linked to how strongly the resonator interacts with the feed line. The area under the resonance dip combines depth and width. Therefore, the feature vector has a physical interpretation rather than being arbitrary [4], [5], [13].

$$f_0 = \frac{1}{2\pi\sqrt{L_{eff}C_{eff}}} \quad (39)$$

$$\frac{\Delta f_0}{f_0} \approx -\frac{1}{2} \frac{\Delta C_{eff}}{C_{eff}} \quad (40)$$

$$C_{eff} = C_0 + C_s(\epsilon_r) \quad (41)$$

$$Q_L = \frac{\omega_0 W_{stored}}{P_{loss} + P_{external}} \quad (42)$$

2.18. Two-Port Resonator Transfer Function and S21 Shape

The exported S21 curve is a frequency-domain transfer function. Around a single isolated resonance, the response can often be approximated by a Lorentzian-like shape or by a rational transfer function. The exact form depends on whether the resonance appears as a transmission peak, transmission dip, notch, or bandpass feature. The current project processes the resonance as a minimum in the exported S21 magnitude, because the MATLAB feature extraction finds the minimum transmission point [4], [5], [13].

The resonance minimum gives f_0 . The 3-dB bandwidth is obtained by finding the frequencies at which the S21 curve rises 3 dB above the minimum level. This convention is suitable for a notch-like resonance. If the resonator is instead implemented as a bandpass peak, the same logic must be modified so that the bandwidth is measured relative to the maximum. The thesis therefore explicitly defines the feature extraction rule used in the MATLAB implementation [4], [5], [13].

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A two-port resonator response is affected by both internal resonator physics and external coupling. If the coupling is weak, the resonance may be shallow and difficult to detect. If the coupling is too strong, the loaded Q may decrease and the resonance may broaden. A useful sensor requires a resonance that is deep enough to track but not so broad that small dielectric changes become invisible [1], [4], [13].

The slope around resonance is important because the slope converts frequency shifts into amplitude changes at a fixed interrogation frequency. A steep slope means that a small frequency shift produces a larger S21 change. This is useful if a practical implementation monitors amplitude at one frequency instead of sweeping the full band. The current model still uses the full sweep, but slope features help encode the local sharpness of the curve [1], [4], [13].

The S21 response should be interpreted in decibels carefully. CST may export magnitude in dB, linear magnitude, or real/imaginary values depending on settings. The MATLAB files include logic to handle common formats. For the final thesis, the exported data should be verified by plotting a known file and confirming that the curve shape matches the CST S21 result window [4], [5], [13].

$$S_{21}(f) = \frac{V_2^-(f)}{V_1^+(f)} \Big|_{V_2^+ = 0} \quad (43)$$

$$D = S_{max} - S_{min} \quad (44)$$

$$BW_{3dB} = f_R - f_L \quad (45)$$

$$Q_L = \frac{f_0}{f_R - f_L} \quad (46)$$

2.19. Cavity Perturbation View of Dielectric Characterization

The perturbation formula gives a field-theoretic explanation of why a resonator can be used as a dielectric sensor. If a small volume of the resonator is perturbed by changing permittivity, the resonant frequency changes in proportion to the electric-field energy stored in that volume. The sign of the shift is negative for an increase in permittivity

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because the effective electrical length or capacitance of the resonant system increases [1], [4], [13].

The numerator of the perturbation expression contains the permittivity change multiplied by the squared electric-field magnitude inside the sample region. This term shows directly why sample placement is critical. A material placed where the electric field is high produces a larger shift. A material placed where the electric field is nearly zero produces almost no effect even if its dielectric constant is large [1], [2].

The denominator contains the total stored energy in the resonator. A resonator with highly localized electric energy in the sample region will have high sensitivity because a larger fraction of the total energy is perturbed. This is one of the reasons CSRR geometries are commonly useful for sensing: their slots and gaps produce concentrated near fields [6]-[8].

The perturbation equation is approximate. It assumes that the field distribution does not change drastically after inserting the sample. For large dielectric changes or large sample volumes, the approximation becomes less exact. Full-wave simulation remains necessary for accurate calibration across a broad epsilon range. The CST sweep effectively computes the nonlinear response without relying only on the small-perturbation assumption [1], [4], [13].

In this thesis, perturbation theory is used as explanatory theory rather than as the only computational model. The actual mapping is generated by CST and learned by MATLAB. This combination is appropriate because the sensing response over $\epsilon_r = 1.0$ to 10.0 can be nonlinear and geometry-specific [4], [5], [13].

$$\frac{\Delta f}{f_0} \approx -\frac{1}{2} \frac{\int_{V_s} (\epsilon_s - \epsilon_0) |\mathbf{E}_0|^2 dV}{\int_V (\epsilon |\mathbf{E}_0|^2 + \mu |\mathbf{H}_0|^2) dV} \quad (47)$$

$$\eta_s = \frac{\int_{V_s} \epsilon |\mathbf{E}|^2 dV}{\int_V \epsilon |\mathbf{E}|^2 dV} \quad (48)$$

$$|S_f| \propto \eta_s \quad (49)$$

2.20. Numerical Frequency Sampling and Resonance Tracking

The extracted resonance frequency is only as reliable as the frequency grid and tracking algorithm. If the simulation sweep has coarse frequency resolution, the minimum point

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may be selected at a nearby grid sample rather than at the true continuous resonance. This creates discretization error. A refined frequency mesh or local interpolation near the resonance can reduce this error [1], [4], [13].

Mode tracking is a second numerical issue. A complex microwave structure may have more than one resonance in the simulated band. If the algorithm simply selects the global minimum across the whole band, it may jump from one mode to another as epsilon changes. Such a jump would produce an artificial discontinuity in the f_0 -epsilon curve. To reduce this risk, the final MATLAB code includes a tracking-window concept that can restrict the search near the previous resonance [4], [5], [13].

The physical expectation is continuity. When epsilon changes by only 0.1, the resonant frequency should normally move by a limited amount, not jump abruptly to a distant frequency. Therefore, the f_0 -epsilon plot is a diagnostic tool. Any large discontinuity must be checked by opening the corresponding S21 curves and confirming whether the same resonance branch was followed [1], [4], [13].

Smoothing is used only to avoid selecting a numerical noise spike as the resonance. It should be mild enough that it does not distort the resonance shape. Excessive smoothing can shift f_0 , alter bandwidth, and reduce the apparent depth. The selected moving-median smoothing in MATLAB is a compromise between noise robustness and shape preservation [4], [5], [13].

A more advanced final implementation could fit a local polynomial, Gaussian, or Lorentzian curve around the resonance. This would estimate f_0 at a continuous frequency rather than at a discrete sample point. However, for the current simulation-based thesis, grid-based extraction with validation plots is sufficient if the frequency resolution is adequate and the resonance trend is physically smooth [1], [4], [13].

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$$f_{0, grid} = \arg \min_f S_{21, dB}(f) \quad (50)$$

$$|f_0(\varepsilon_k) - f_0(\varepsilon_{k-1})| < \Delta f_{track} \quad (51)$$

$$\hat{f}_0 = \arg \min_f p_2(f) \quad (52)$$

2.21. Physical Meaning of Extracted Features and Feature Redundancy

Each extracted feature represents a different projection of the S21 curve. F0_GHz represents the horizontal movement of the resonance. Smin_dB represents the strongest attenuation or weakest transmission point. Smax_dB provides a reference level within the search band. Depth_dB measures resonance contrast. Bandwidth3dB_GHz and LoadedQ represent frequency selectivity. AreaBelowMax combines depth and width into an integral descriptor [1], [4], [13].

Feature redundancy is not necessarily harmful for tree ensembles. Some features are mathematically related, such as bandwidth and Q. However, redundancy can still be useful when one feature is noisy or when the relationship is nonlinear. For example, Q depends on f0 and bandwidth, but presenting both f0 and Q allows the model to split on either variable directly. The final feature table should still be inspected for abnormal values [3], [5], [11], [14].

Correlation between features is expected because they originate from the same physical curve. A high correlation does not automatically mean a feature should be removed. The decision should be based on validation performance, physical meaning, and robustness. For a future publication-level study, feature importance and ablation analysis can be added to determine which S21 descriptors contribute most to epsilon prediction [1], [4], [13].

The multi-feature approach is especially important if the project is extended from purely real epsilon to complex permittivity. The real part of permittivity mainly shifts the resonance, while the imaginary part can affect depth and bandwidth. Therefore, the current architecture is already compatible with future multi-output models that estimate both epsilon prime and epsilon double-prime [1], [4], [13].

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$$A = \int_{f_{min}}^{f_{max}} (S_{max} - S_{21, dB}(f)) df \quad (53)$$

$$m_L = \left. \frac{dS_{21, dB}}{df} \right|_{f < f_0}, \quad m_R = \left. \frac{dS_{21, dB}}{df} \right|_{f > f_0} \quad (54)$$

2.22. Regression Tree Mathematics in Greater Detail

A regression tree partitions the feature space into rectangular regions. At each internal node, the algorithm selects one feature and one threshold. Samples with feature values below the threshold go to one branch, and the remaining samples go to the other branch. The selected split is the one that reduces target variance most effectively. For this thesis, the target is ϵ_r , so each split tries to create child regions with more homogeneous ϵ_r values [3], [5], [11], [14].

At a terminal leaf, the tree prediction is usually the mean target value of training samples falling in that region. This produces a piecewise-constant approximation to the calibration function. A single tree can overfit if it grows too deep, because it may create leaves that describe only one or two training samples. Tree depth, minimum leaf size, and ensemble averaging are therefore important [3], [5], [11], [14].

Bagging reduces variance by training many trees on bootstrap samples. A bootstrap sample is generated by sampling the training set with replacement. Each tree sees a slightly different dataset. The final prediction is the average of the individual tree predictions. This average is more stable than a single tree because unstable split decisions are partially cancelled across the ensemble [3], [5], [11], [14].

Random Forest usually adds another randomization step by selecting a subset of candidate features at each split. MATLAB bagged regression trees may be configured in ways that are closely related to Random Forest behavior. In this thesis, the model is described as a bagged regression-tree or Random-Forest-like ensemble because the central idea is ensemble averaging of tree predictors [3], [5], [11], [14].

The advantage of a tree ensemble in this project is that it can learn nonlinear calibration behavior without requiring an explicit polynomial model. The disadvantage is limited extrapolation. If a measured S21 curve lies outside the feature range created by ϵ_r 1 to 10, the prediction may not be reliable. This is why the trained interface should be used within the calibration interval [3], [5], [11], [14].

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$$R_m = \{\mathbf{x} : x_j \leq s\} \quad (55)$$

$$\hat{y}_{leaf} = \frac{1}{N_{leaf}} \sum_{i \in leaf} y_i \quad (56)$$

$$\mathcal{B}_b = \{(\mathbf{x}_{i_b}, y_{i_b})\}_{i=1}^N \quad (57)$$

$$\text{Var} \left[\frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}) \right] \leq \text{Var}[T(\mathbf{x})] \quad (58)$$

2.23. Bias-Variance Interpretation of the Learning Model

The prediction error of a supervised regression model can be interpreted through bias and variance. Bias describes systematic error caused by an overly simple model. Variance describes sensitivity to the particular training dataset. A shallow model may have high bias because it cannot represent nonlinear calibration behavior. A very flexible model may have high variance because it adapts too strongly to individual samples [3], [5], [11], [14].

The microwave calibration problem can be nonlinear because the resonant frequency shift is not necessarily linear with epsilon. The perturbation approximation predicts a nearly linear shift only for small perturbations. Across epsilon values from 1 to 10, the field distribution and coupling may change enough to create nonlinear behavior. Therefore, a model with enough flexibility is needed [1], [2].

Bagging is useful because regression trees are flexible but unstable. Small changes in the dataset can lead to different split thresholds. Averaging many trees reduces variance without forcing the calibration function to be globally linear. This is suitable for a simulation-generated dataset where the number of samples is moderate and the feature-target relationship may be nonlinear [3], [5], [11], [14].

Cross-validation estimates how well the model generalizes to unseen epsilon cases within the simulated calibration interval. It does not prove that the model will work on real measured liquids. Experimental generalization requires measured validation data. Therefore, the thesis distinguishes numerical validation from physical validation [3], [5], [11], [14].

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$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2 \quad (59)$$

$$\hat{f}_{bag}(\mathbf{x}) = \mathbb{E}_B[T(\mathbf{x}; B)] \quad (60)$$

2.24. Cross-Validation Design for the 91-Point Dataset

The final dataset generated by $\epsilon_r = 1.0$ to 10.0 with 0.1 spacing contains 91 simulated samples. This is larger than the initial 19-point dataset and provides a denser calibration curve. However, it is still a structured one-dimensional sweep in terms of the underlying physical parameter. Therefore, validation must be interpreted carefully. Random k-fold cross-validation tests interpolation between simulated cases, not extrapolation to unknown physical conditions [3], [5], [11], [14].

A useful additional check is leave-block-out validation. Instead of randomly removing individual samples, contiguous ϵ ranges can be withheld. This tests whether the model can interpolate across a missing interval. Although the current MATLAB file uses k-fold or leave-one-out validation depending on dataset size, the thesis can mention block validation as a future improvement for more rigorous assessment [4], [5], [13].

The predicted-versus-true plot should be accompanied by an error-versus- ϵ plot. A single RMSE value can hide local weaknesses. If the error is low in the middle of the range but high near $\epsilon = 10$, the model may have boundary problems. If the error spikes at one ϵ value, the corresponding S21 curve may contain tracking error or export failure [3], [5], [11], [14].

For a publication-quality final model, the dataset can be divided into training, validation, and test sets. The validation set would guide model settings such as tree number and minimum leaf size. The test set would be used only once for final reporting. In the current graduation-project workflow, cross-validation is a practical and acceptable evaluation method for the simulation stage [3], [5], [11], [14].

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$$RMSE_{CV} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_{i, -fold})^2} \quad (61)$$

$$e_i = \hat{\varepsilon}_i - \varepsilon_i \quad (62)$$

$$\bar{e} = \frac{1}{N} \sum_{i=1}^N e_i \quad (63)$$

2.25. Data Integrity, Naming Convention, and Automation Reliability

The automation stage is not only a convenience; it is part of the scientific method of the thesis. Manual CST simulation of 91 cases would be slow and error-prone. The MATLAB-CST automation ensures that each case is generated by the same solver process, the same export rule, and the same naming convention. This improves reproducibility and reduces accidental differences between files [4], [5], [13].

The file name `eps_%.2f.txt` encodes the target epsilon value. The model-training script parses this value from the file name and uses it as the supervised-learning target. If a file is misnamed, the model will learn an incorrect label. Therefore, the export log and feature table are important quality-control documents. They make it possible to trace each model row back to a specific CST simulation [4], [5], [13].

The corrected automation exports S2,1, not S1,1. This correction is critical because the thesis is built around the two-port transmission response. If S11 were exported while the text described S21, the physical interpretation and model features would be inconsistent. The revised script eliminates this inconsistency by using the CST tree item corresponding to S-Parameters S2,1 [4], [5], [13].

The automation also checks whether export files exist and contain data. This prevents an empty or failed file from silently entering the dataset. In future work, additional checks can compare frequency range, number of samples, and minimum/maximum S21 values across all files to detect abnormal exports automatically [4], [5], [13].

2.26. Relationship Between Simulation and Future VNA Measurement

A physical implementation would measure S21 with a vector network analyzer. The VNA sends a known microwave signal into port 1 and measures the transmitted wave at port 2. Calibration is required to remove systematic errors due to cables, connectors,

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adapters, and fixture transitions. Without calibration, the measured S21 curve includes both the sensor response and the measurement setup response [1], [4], [13].

The simulation assumes idealized geometry, defined material properties, and numerical boundary conditions. A fabricated PCB will include manufacturing tolerance, copper roughness, solder effects, connector parasitics, substrate-property variation, and sample-placement uncertainty. Therefore, measured S21 will not exactly match CST. This does not make the simulation useless; it means the simulation should be considered the first calibration layer [1], [4], [13].

The next development step is to measure reference materials with known dielectric constants and compare their measured resonances with simulated resonances. Differences can be used to correct the CST model or retrain the machine-learning model with measured data. A combined simulation-measurement dataset would improve practical reliability [1], [4], [13].

Temperature control is also necessary for liquids. Many liquids have temperature-dependent permittivity and loss tangent. If the same liquid is measured at different temperatures, the resonance may shift even without concentration change. A future model can include temperature as an additional input feature. The present thesis identifies this factor as a limitation rather than ignoring it [1], [2].

$$S_{21, meas}(f) = S_{setup}(f)S_{sensor}(f) \quad (64)$$

$$S_{21, cal}(f) = \frac{S_{21, meas}(f)}{S_{setup}(f)} \quad (65)$$

$$\hat{\epsilon} = g(\mathbf{x}, T) \quad (66)$$

2.27. Summary of the Expanded Mathematical Framework

The complete mathematical framework connects electromagnetic field theory, resonator physics, signal features, and supervised learning. Maxwell equations describe how fields interact with materials. Complex permittivity describes energy storage and loss. Resonator theory explains the frequency-selective response. Perturbation theory explains why dielectric loading shifts resonance. S-parameters define the measurable two-port response. Feature extraction converts the S21 curve into numerical descriptors. Regression learns the inverse mapping from descriptors to epsilon [4], [5], [13].

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The final results chapter will close the loop by showing whether the simulated S21 curves behave consistently with theory. If epsilon increases and the resonance shifts in the expected direction, the physical model is supported. If validation metrics show low error, the learning model is supported. If both occur together, the thesis demonstrates a coherent AI-enhanced microwave dielectric sensor workflow [1], [4], [13].

After these theoretical developments, the final results chapter can be written in a technically precise way. Each curve, table, and validation metric will be connected to one of the mathematical concepts above: frequency shift to perturbation theory, bandwidth to loaded Q, depth to coupling, error metrics to supervised regression, and automation outputs to reproducibility [4], [5], [13].

2.28. Analytical Approximation of the Calibration Curve

The CST-generated calibration curve can be interpreted analytically by relating the resonant frequency to an effective capacitance. If the resonator is approximated by an effective inductance and capacitance, the sample-dependent part of capacitance can be written as a weighted function of relative permittivity. For small perturbations this dependence may be approximately linear, but across the wide interval from $\epsilon_r = 1$ to 10 the response can become nonlinear because the field distribution and coupling condition also change [1], [4], [13].

A first-order model assumes that the sample contribution is proportional to ϵ_r minus the reference value. This gives a rational square-root relationship between f_0 and ϵ_r . The relationship is often monotonic but not exactly linear. The Random Forest model does not require this analytical expression, but the expression is useful for checking whether the simulated curve has a physically reasonable direction and curvature [3], [5], [11], [14].

If the f_0 -epsilon plot is smooth and decreasing, the resonator behaves consistently with capacitive dielectric loading. If the curve has a strong local reversal, the engineer should check whether another resonance branch was selected. If the curve saturates at high epsilon, this can be explained by the fact that only a finite fraction of the total electric field energy is located in the sample volume [1], [2].

The analytical approximation also clarifies the limit of sensitivity. If most of the electric field is outside the sample region, the coefficient multiplying ϵ_r will be small and the resonance shift will be weak. If a large field fraction overlaps the sample region, the same epsilon change causes a larger shift. This explains why sensor geometry and sample placement are as important as the regression model itself [1], [4], [13].

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$$C_{eff}(\epsilon_r) = C_b + k_c(\epsilon_r - \epsilon_{ref}) \quad (68)$$

$$f_0(\epsilon_r) = \frac{1}{2\pi\sqrt{L_{eff}[C_b + k_c(\epsilon_r - \epsilon_{ref})]}} \quad (69)$$

$$\frac{df_0}{d\epsilon_r} = -\frac{k_c}{4\pi\sqrt{L_{eff}[C_{eff}(\epsilon_r)]^{3/2}}} \quad (70)$$

2.29. CST Solver, Port, Boundary, and Mesh Considerations

The reliability of the simulated S21 curves depends on the electromagnetic solver setup. Port definition determines how waves enter and leave the structure. If the ports are not assigned correctly, the exported S2,1 response may include artificial mismatch or may not represent the intended two-port path. The port reference impedance should be consistent with the microwave interpretation of the sensor, typically 50 ohms for standard RF systems [1], [4], [13].

Boundary conditions determine how the finite simulation domain represents open space or shielding. If boundaries are too close to the resonator, they may alter the fields and shift the resonance. Open or absorbing boundaries should be placed sufficiently far away when radiation is possible. In planar sensor simulations, the substrate, ground plane, conductor thickness, and sample volume must be represented consistently so that the field distribution is physically meaningful [1], [4], [13].

Mesh density affects resonator accuracy because slots, gaps, and sharp conductor edges concentrate fields. A coarse mesh can shift the resonance frequency and underestimate local field intensity. A very fine mesh improves accuracy but increases memory and simulation time. The selected workflow balances computational cost and dataset size: the epsilon sweep contains 91 simulations, so each simulation must be accurate enough but still feasible on the available computer [1], [4], [13].

A practical verification step is to run a few selected epsilon values with a refined mesh and compare their resonance frequencies with the normal-mesh results. If the difference is small compared with the shift caused by epsilon changes, the mesh is acceptable for the calibration study. If the difference is large, mesh refinement is necessary before trusting the machine-learning dataset [1], [4], [13].

The CST tree item used for export is also a technical risk point. The automation script selects the S-parameter path corresponding to S2,1. If the CST project stores the response

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under a different internal name, the script must be adjusted to match the actual result tree. This is not a theoretical weakness of the method; it is an implementation detail of the CST automation interface [4], [5], [13].

$$Z_{ref} = 50 \, \Omega \quad (71)$$

$$\Delta f_{mesh} = |f_{0, fine} - f_{0, normal}| \quad (72)$$

2.30. Detailed MATLAB Data Pipeline and Quality-Control Logic

The MATLAB side of the project performs three different functions: control, signal processing, and learning. In the control stage, MATLAB communicates with CST and creates the dataset. In the signal-processing stage, MATLAB reads each S21 file, normalizes the frequency units, smooths the curve if necessary, locates the resonance, and calculates numerical descriptors. In the learning stage, MATLAB trains the ensemble model and evaluates it with cross-validation [3], [5], [11], [14].

Data quality control starts before learning. Each file must have at least two numerical columns and enough frequency samples. The frequency column must be monotonic after sorting. The S21 column must be finite. Empty files, corrupted files, and files with missing columns should be rejected. This prevents the model from learning from invalid curves. The final code performs these checks and skips invalid files with warnings [4], [5], [13].

A second quality check is physical trend inspection. The script creates S21 family plots and f_0 -epsilon plots. These are not decorative figures; they are validation tools. A machine-learning metric alone cannot reveal whether the resonance branch was physically consistent. Therefore, the thesis includes graphical checks as part of the method [1], [4], [13].

The extracted feature table is one of the most important intermediate outputs. It transforms raw S21 curves into an engineering dataset. Each row contains file name, target epsilon, and physically interpretable features. Saving this table as CSV allows the results to be inspected outside MATLAB, imported into other software, or reused in the thesis without rerunning CST [4], [5], [13].

The model file stores the trained model, feature names, metrics, and data table. This is important because the model deployment interface must use the same feature ordering as the model-training script. If feature order changes between training and prediction, the

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numerical prediction becomes meaningless. The code avoids this by storing and reusing the featureNames variable [1], [4], [13].

$$\mathbf{X} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} \in \mathbb{R}^N \quad (73)$$

$$p = 12, \quad N = 91 \quad (74)$$

2.31. Defense-Oriented Interpretation of the Project Contribution

The project should not be described as a device that can identify any liquid with certainty. That claim would be scientifically weak because many liquids can share similar dielectric constants, and real identification requires additional information. The correct claim is that the system estimates effective relative dielectric constant from the S21 response of a calibrated resonator within a defined epsilon interval [1], [4], [13].

A likely question in the defense is why artificial intelligence is needed if f_0 already shifts with epsilon. The answer is that f_0 is only one descriptor and may be insufficient when loss, coupling, bandwidth, and curve shape change. The machine-learning model combines several physically meaningful descriptors and learns a nonlinear calibration function. This makes the system more flexible than a single linear fit [1], [2].

Another likely question is whether simulation results are enough. The correct answer is that this thesis is a proof-of-concept and numerical calibration study. It establishes the design and computational pipeline. Physical fabrication and VNA validation are the next step. This limitation is explicitly stated, which makes the thesis scientifically honest and technically defensible [1], [4], [13].

$$\hat{\varepsilon} = g(f_0, S_{min}, D, BW, Q_L, \Delta f, A, m_L, m_R, \dots) \quad (75)$$

The theoretical chapter is now sufficiently detailed to support the later results section. When the final CST and MATLAB outputs are available, the results will not be presented

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as isolated plots; they will be interpreted through the field, circuit, S-parameter, and machine-learning relationships developed above [4], [5], [13].

3. RESULTS AND DISCUSSIONS

This chapter presents the final outputs of the CST-MATLAB workflow and interprets them in relation to the electromagnetic theory developed in Chapter 2. The final dataset generation stage completed 91 individual CST simulations for ϵ_r values from 1.0 to 10.0 with 0.1 spacing. No failed jobs were reported. The subsequent MATLAB model-training stage extracted twelve S21-based features and evaluated a bagged regression-tree model using 5-fold cross-validation [4], [5].

3.1. CST Geometry, Parameterization, and Port Configuration

The CST model consists of a CSRR-based resonant sensing region, a parameterized dielectric loading block, and two waveguide ports. The sample permittivity is controlled by the parameter ϵ_{degeri} . During the automated sweep, only this dielectric parameter is changed; the sensor geometry, port locations, frequency range, and solver configuration remain fixed. This makes the generated dataset a controlled electromagnetic experiment [4], [13].

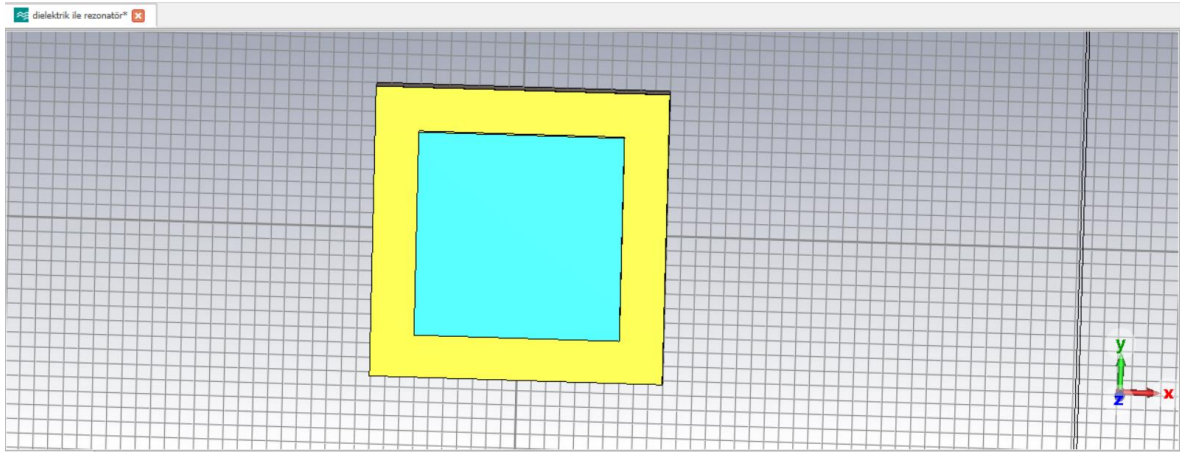


Figure 3.1. Two-port CST sensor geometry view showing dielectric loading region and waveguide-port placement in the simulated structure.

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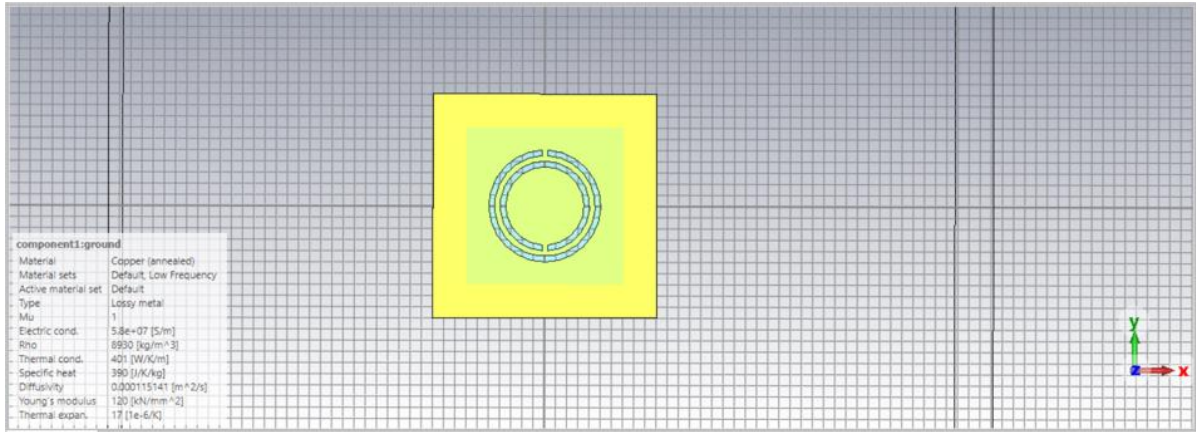


Figure 3.2. Original CST front-view screenshot of the CSRR/circular resonator region and dielectric loading area used for dielectric sensitivity analysis.

3.2. Reference Impedance and CST Port-Quantity Verification

Before the final dataset was generated, the S-parameters were normalized to the standard RF reference impedance of 50 ohms. This correction is important because the early port result indicated a different computed impedance; reporting a 50 ohm microwave system while exporting non-normalized data would create an inconsistency. The final simulation run was performed after 50 ohm normalization. Additional CST result quantities including VSWR, Y-parameters, and Z-parameters were inspected to verify the port behavior and resonant nature of the structure [1], [4], [13].

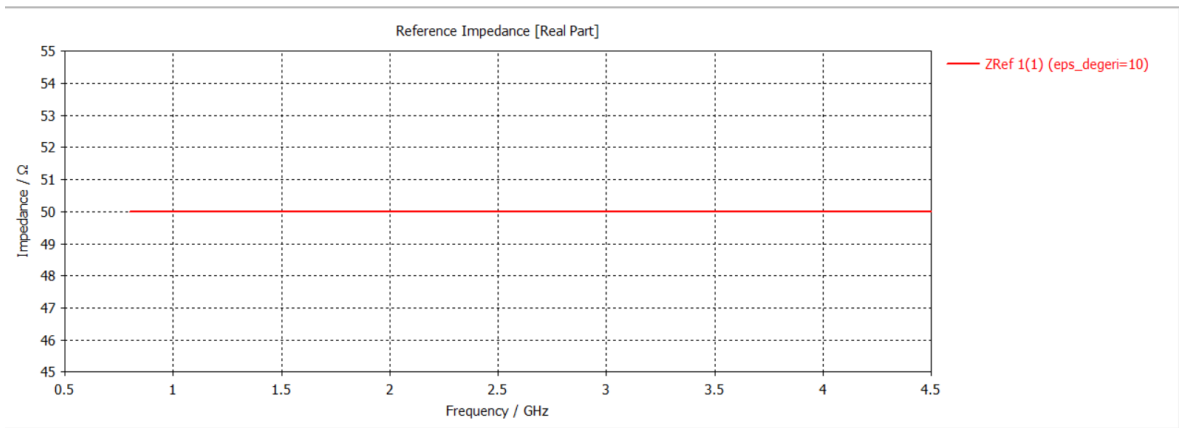


Figure 3.3. Reference impedance verification showing constant 50 ohm normalization over the simulated frequency range.

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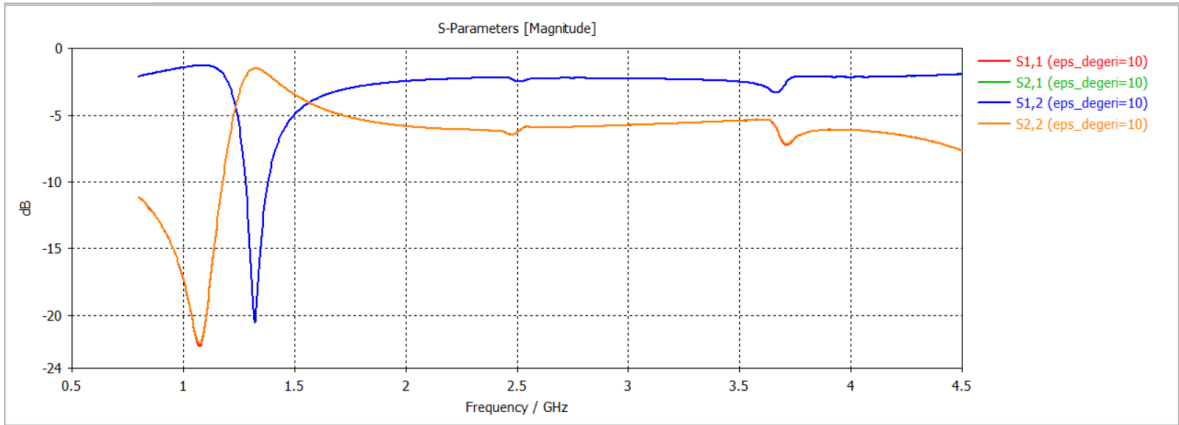


Figure 3.4. CST S-parameter magnitude response including S11, S21, S12, and S22 after the final normalization workflow.

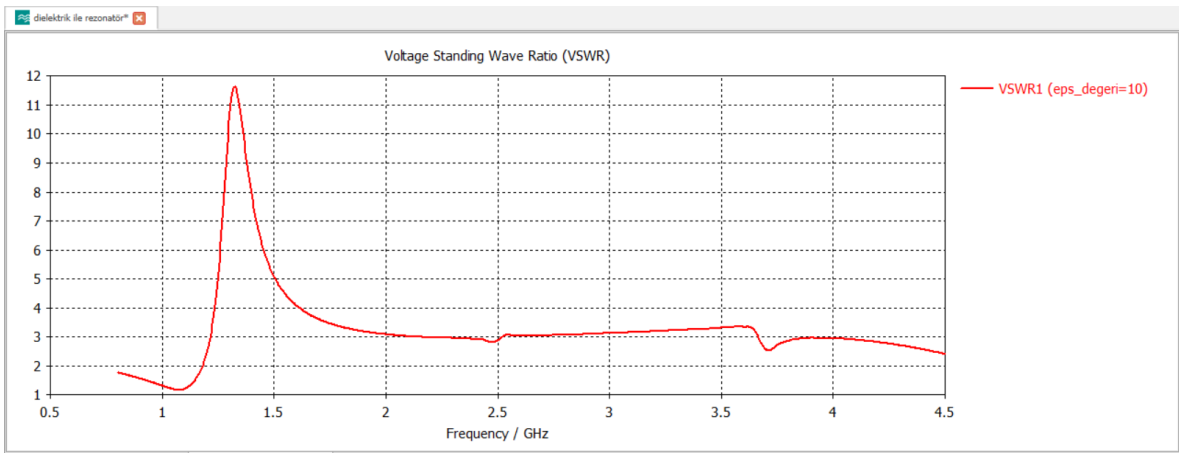


Figure 3.5. VSWR response inspected after CST simulation as an additional port-quality and resonance-indication output.

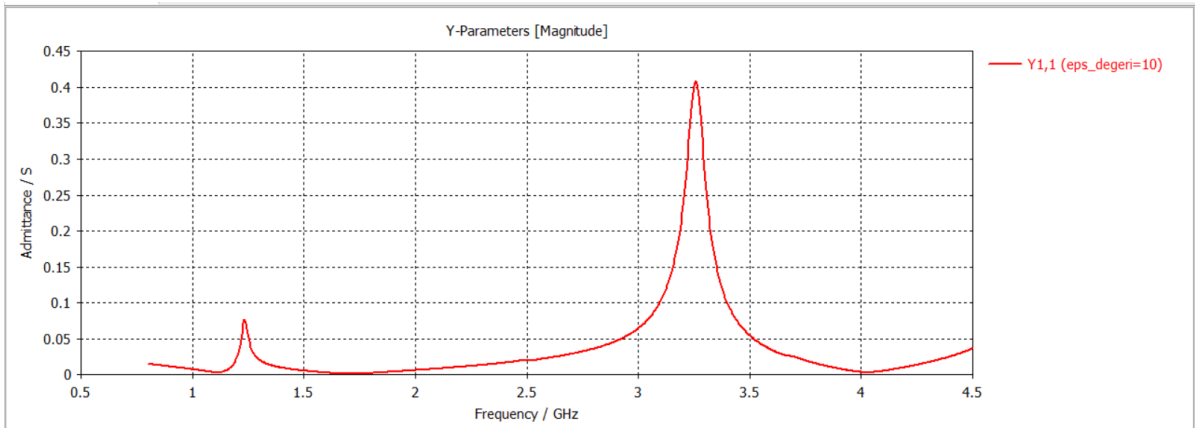


Figure 3.6. Y-parameter magnitude response used as an admittance-domain verification of resonant behavior.

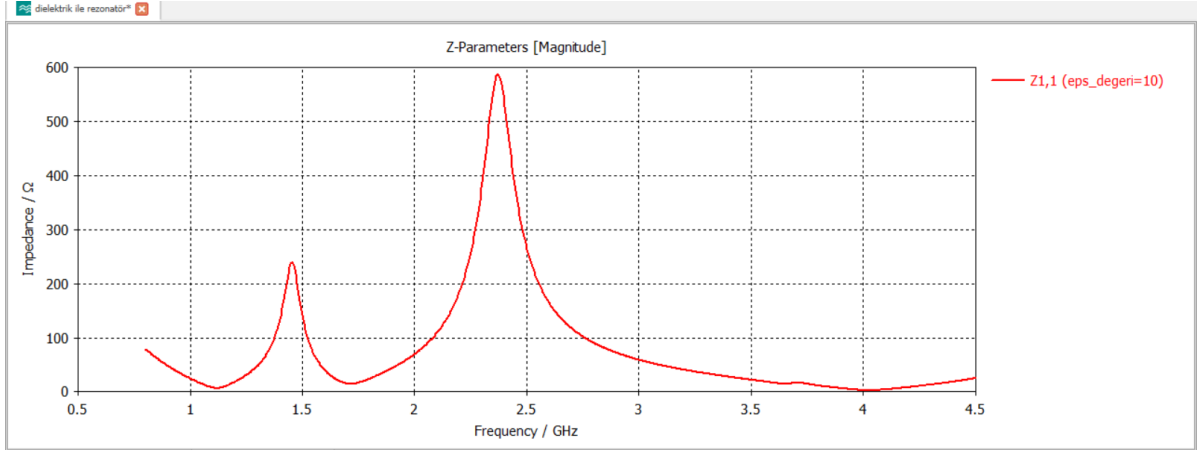


Figure 3.7. Z-parameter magnitude response used as an impedance-domain verification of resonant behavior.

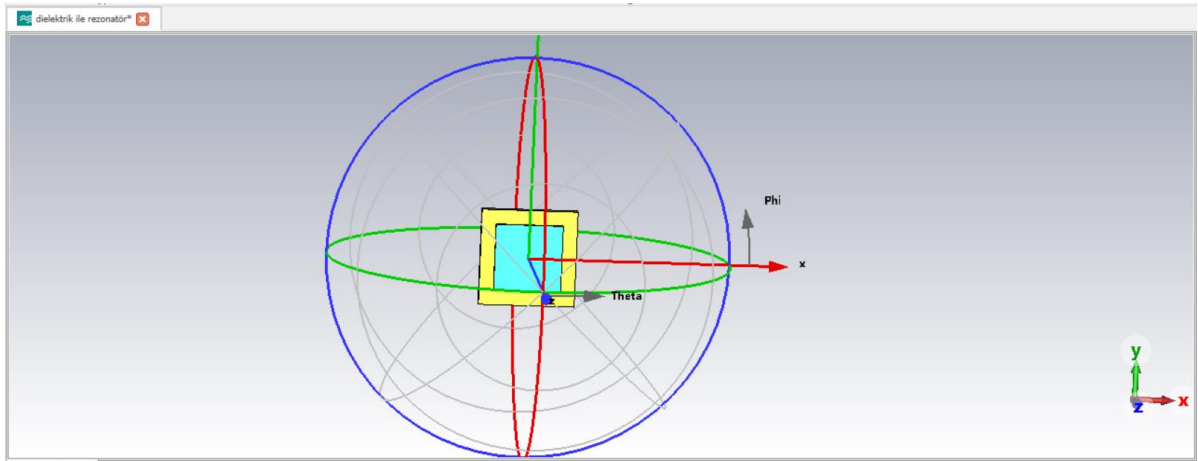


Figure 3.8. CST angular reference visualization used to inspect coordinate orientation and field-plot interpretation.

3.3. CST Dataset Generation and Quality Control

The final CST-MATLAB automation completed the dielectric sweep from $\epsilon_r = 1.0$ to 10.0 . The step size was 0.1 , producing 91 exported S21 text files. The export log reported 91 successful files and 0 failed jobs. This is a central reproducibility result because every subsequent feature and model output depends on the completeness and consistency of these S21 files [5], [13].

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eps_7.30	6.06.2026 20:08	Metin Belgesi	57 KB
eps_7.40	6.06.2026 20:12	Metin Belgesi	57 KB
eps_7.50	6.06.2026 20:15	Metin Belgesi	57 KB
eps_7.60	6.06.2026 20:19	Metin Belgesi	57 KB
eps_7.70	6.06.2026 20:23	Metin Belgesi	57 KB
eps_7.80	6.06.2026 20:27	Metin Belgesi	57 KB
eps_7.90	6.06.2026 20:32	Metin Belgesi	57 KB
eps_8.00	6.06.2026 20:36	Metin Belgesi	57 KB
eps_8.10	6.06.2026 20:40	Metin Belgesi	57 KB
eps_8.20	6.06.2026 20:44	Metin Belgesi	57 KB
eps_8.30	6.06.2026 20:48	Metin Belgesi	57 KB
eps_8.40	6.06.2026 20:52	Metin Belgesi	57 KB
eps_8.50	6.06.2026 20:55	Metin Belgesi	57 KB
eps_8.60	6.06.2026 20:58	Metin Belgesi	57 KB
eps_8.70	6.06.2026 21:02	Metin Belgesi	57 KB
eps_8.80	6.06.2026 21:05	Metin Belgesi	57 KB
eps_8.90	6.06.2026 21:08	Metin Belgesi	57 KB
eps_9.00	6.06.2026 21:12	Metin Belgesi	57 KB
eps_9.10	6.06.2026 21:15	Metin Belgesi	57 KB
eps_9.20	6.06.2026 21:18	Metin Belgesi	57 KB
eps_9.30	6.06.2026 21:22	Metin Belgesi	57 KB
eps_9.40	6.06.2026 21:25	Metin Belgesi	57 KB
eps_9.50	6.06.2026 21:29	Metin Belgesi	57 KB
eps_9.60	6.06.2026 21:32	Metin Belgesi	57 KB
eps_9.70	6.06.2026 21:35	Metin Belgesi	57 KB
eps_9.80	6.06.2026 21:39	Metin Belgesi	57 KB
eps_9.90	6.06.2026 21:43	Metin Belgesi	57 KB
eps_10.00	6.06.2026 21:47	Metin Belgesi	57 KB

Figure 3.9. Original S21_Exports folder output showing the final eps_*.txt files generated by the dielectric sweep up to eps_10.00.txt.

Table 3.1. Final CST dataset generation settings and quality-control summary.

Item	Final value
Project root	C:\Users\CEM\Desktop\Bitirme2
CST model	dielektrik ile rezonatör.cst
Swept parameter	eps_degeri
Sweep interval	epsilon_r = 1.0 to 10.0
Step size	0.1
Number of generated S21 files	91
Failed jobs	0
Exported response	S21 / S2,1
Reference impedance	50 ohm

3.4. S21 Family Curves for the Dielectric Sweep

Figure 3.10 shows the complete family of S21 curves generated from the 91-point epsilon sweep. The dominant notch shifts from the higher-frequency side toward the lower-

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frequency side as dielectric constant increases. This is exactly the behavior expected from dielectric loading: higher permittivity increases effective capacitance and lowers the resonance frequency. The notch remains deep and trackable across the sweep, which confirms that the selected resonant feature can be used as the main sensing signature [1], [7], [8].

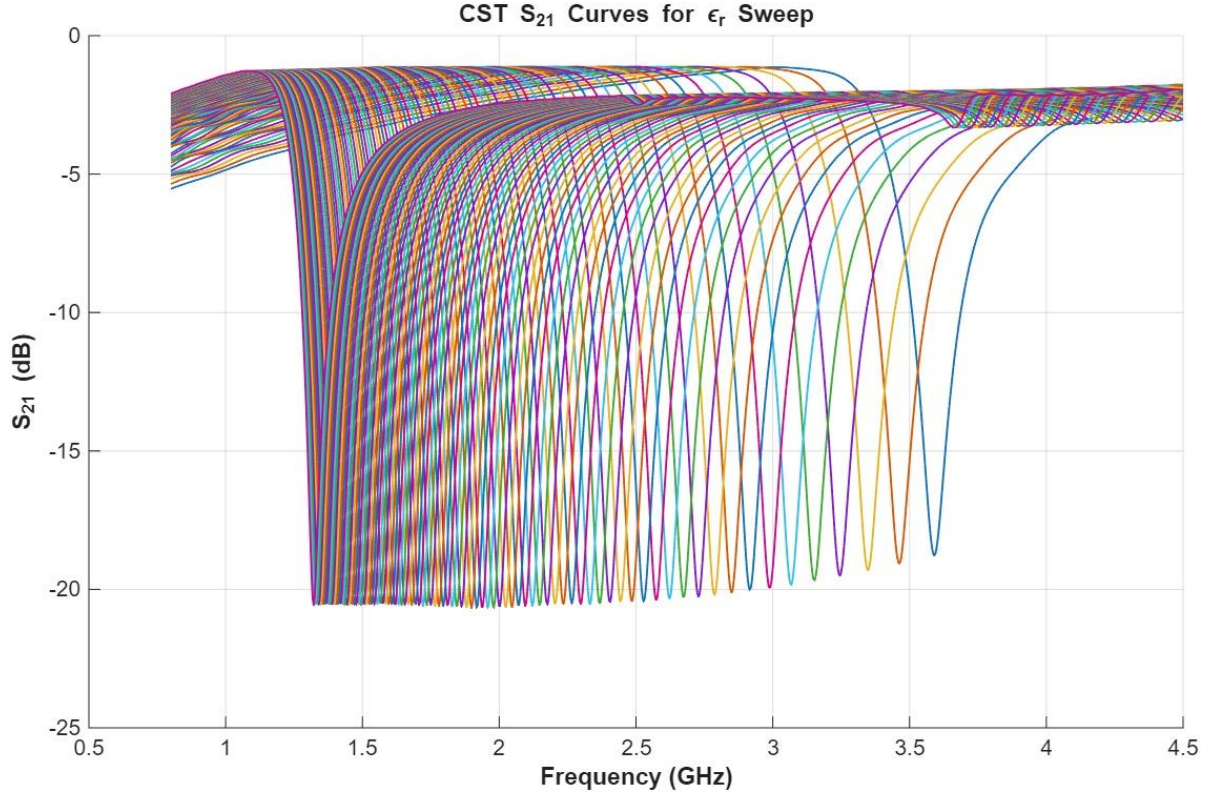


Figure 3.10. CST-generated S21 transmission curves for the epsilon_r sweep from 1.0 to 10.0.

3.5. Resonance Frequency and Dielectric Constant Calibration

The extracted resonance-frequency calibration curve is shown in Figure 3.11. The result is smooth, monotonic, and nonlinear. At epsilon_r = 1.0, the resonance frequency is approximately 3.58 GHz. At epsilon_r = 10.0, the resonance frequency decreases to approximately 1.31 GHz. The total frequency shift is therefore about 2.27 GHz. This large shift demonstrates strong dielectric sensitivity in the simulated structure [1], [7], [8].

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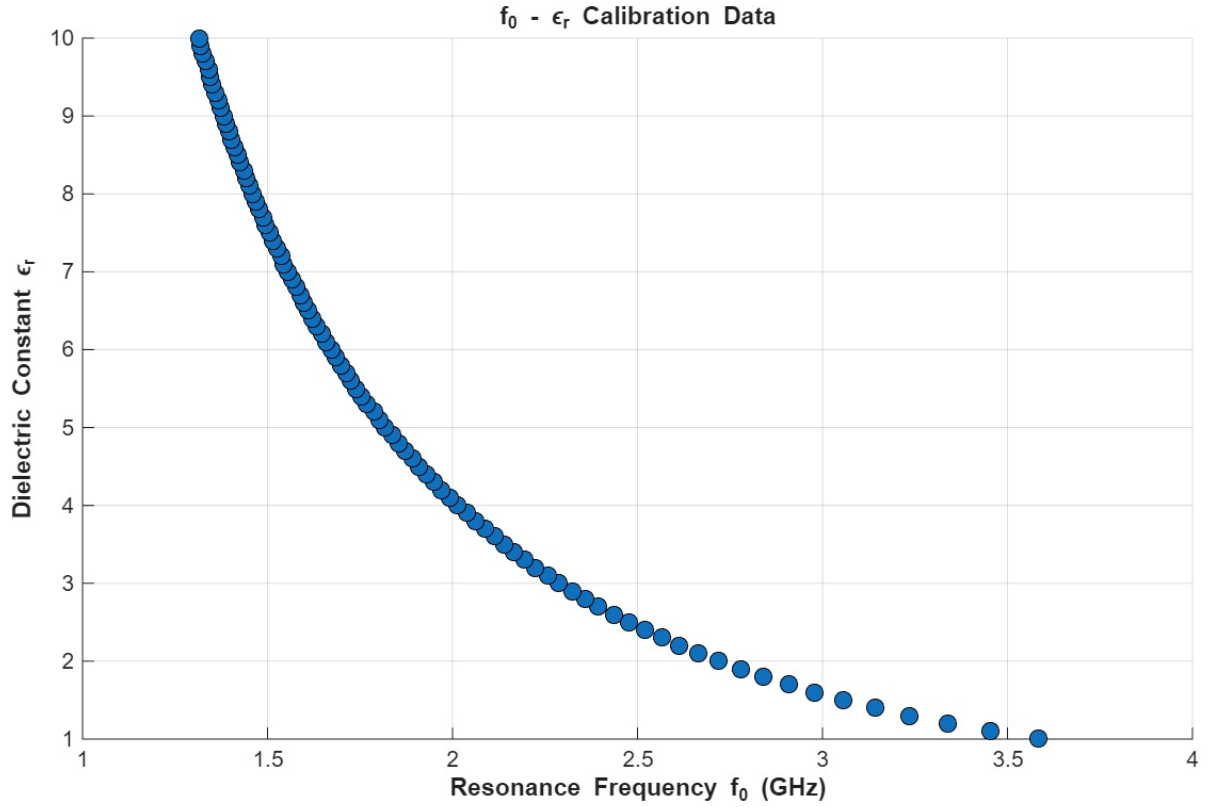


Figure 3.11. Resonance frequency versus dielectric constant calibration data extracted from the S21 curves.

Table 3.2. Summary statistics of selected extracted S21 features.

Feature	Observed range / value
Number of samples	91
Epsilon range	1.0 to 10.0
F0 GHz	3.5824 GHz down to 1.3143 GHz
Smin dB	approximately -18.36 dB to -19.76 dB
Depth dB	approximately 17.22 dB to 18.64 dB
Bandwidth3dB GHz	approximately 0.0444 GHz to 0.0777 GHz
LoadedQ	approximately 27.32 to 47.48
Maximum DeltaF GHz	approximately -2.2681 GHz

3.6. Extracted Feature Dataset

The feature table transforms raw S21 curves into a supervised learning dataset. Each row contains the target dielectric constant and twelve physically meaningful features. These features represent resonance position, resonance depth, bandwidth, loaded Q, frequency shift, area, mean response, standard deviation, and local slopes. The model therefore learns from the resonance shape rather than from a single scalar f0 value alone [1], [4], [13].

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Table 3.3. Representative rows from the extracted feature table.

Epsilon	F0 GHz	Smin dB	Depth dB	Bandwidth3dB GHz	LoadedQ	DeltaF GHz
1.0	3.5824	-18.36	17.22	0.0777	46.11	0.0000
3.2	2.2245	-19.69	18.56	0.0555	40.08	-1.3579
5.5	1.7398	-19.34	18.15	0.0518	33.59	-1.8426
7.8	1.4771	-19.07	17.84	0.0444	33.27	-2.1053
10.0	1.3143	-18.56	17.28	0.0481	27.32	-2.2681

3.7. Random Forest Regression Performance

The final model was evaluated using 5-fold cross-validation. The regression results show very strong numerical interpolation performance within the simulated dielectric interval. The R-squared value of 0.9967 indicates that the model explains nearly all of the variance in the simulated epsilon labels. The RMSE of approximately 0.150 epsilon and MAE of approximately 0.105 epsilon show that the average prediction error is small compared with the full epsilon interval from 1 to 10 [3], [5], [11].

```

===== DOGRULAMA SONUCLARI =====
Model tipi      : Bagged Regression Trees / Random Forest benzeri ensemble
Veri sayisi     : 91
Ozellik sayisi  : 12
CV yontemi      : 5-Fold Cross Validation
RMSE            : 0.150374 epsilon
MAE             : 0.104530 epsilon

```

Figure 3.12. MATLAB command-window validation output for the final 91-sample dataset and 12-feature Random Forest regression model.

Table 3.4. Cross-validation performance metrics of the regression model.

Metric	Value
Validation method	5-Fold Cross Validation
Number of samples	91
Number of features	12
RMSE	0.150374 epsilon
MAE	0.104530 epsilon
R-squared	0.996723
Maximum absolute error	0.541632 epsilon

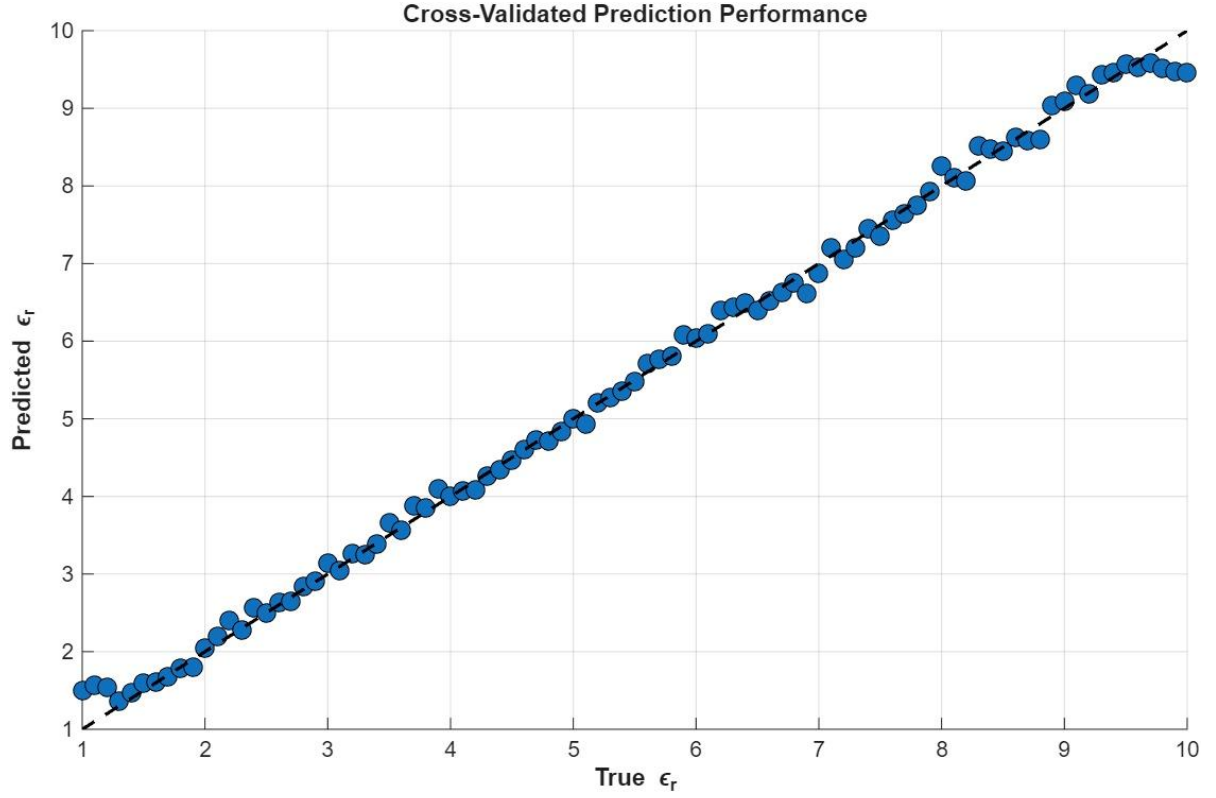


Figure 3.13. Cross-validated predicted versus true dielectric constant values.

3.8. Cross-Validation Error Distribution

The cross-validation error distribution is centered around zero, showing that the model does not exhibit a strong overall positive or negative bias. Most errors remain within a narrow band, while the largest deviations occur near the boundaries of the calibration interval. This boundary behavior is expected for tree-based models because predictions near the edge of the training range are constrained by fewer neighboring samples [3], [5], [11], [14].

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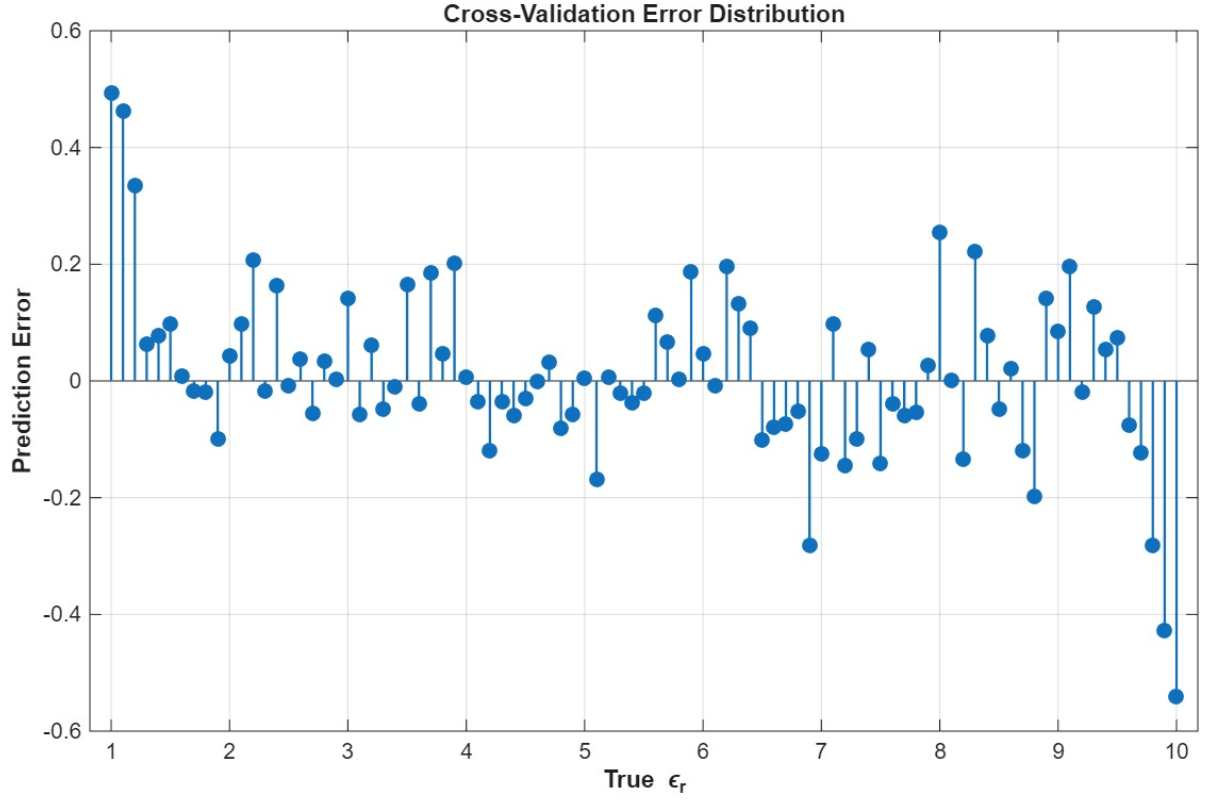


Figure 3.14. Cross-validation error distribution as a function of true dielectric constant.

3.9. Learned Calibration Trend

Figure 3.15 compares the trained Random Forest calibration trend with the CST-derived calibration data. The model follows the nonlinear inverse relation between f_0 and ϵ_r closely. This confirms that the selected S21 features are sufficient for constructing a numerical inverse mapping from resonator response to effective dielectric constant in the simulated interval.

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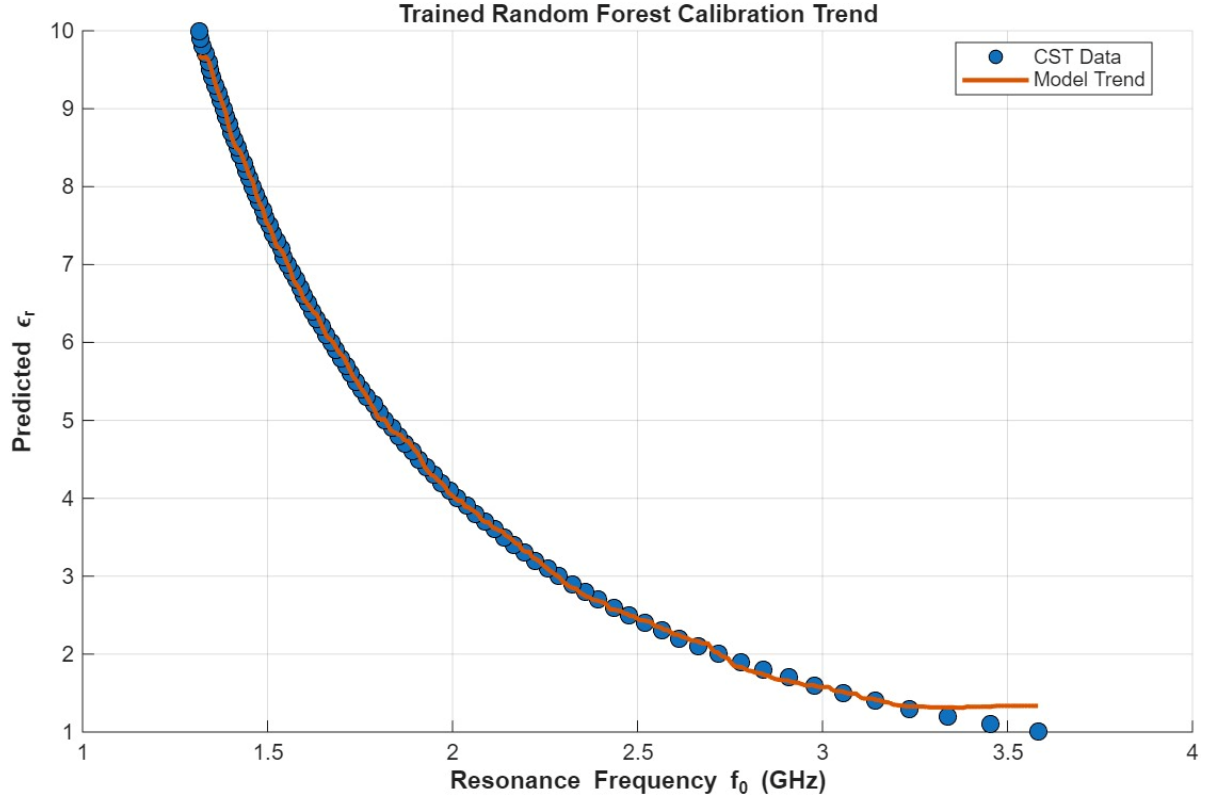


Figure 3.15. Trained Random Forest calibration trend compared with CST calibration data.

The optional MATLAB interface was prepared only as a visualization and demonstration layer. It displays the calibration curve, representative CST S21 curves, cross-validation summary, and an approximate prediction field. It is not used to claim direct chemical identification; the underlying model estimates the effective dielectric constant from S21-derived information [5], [11].

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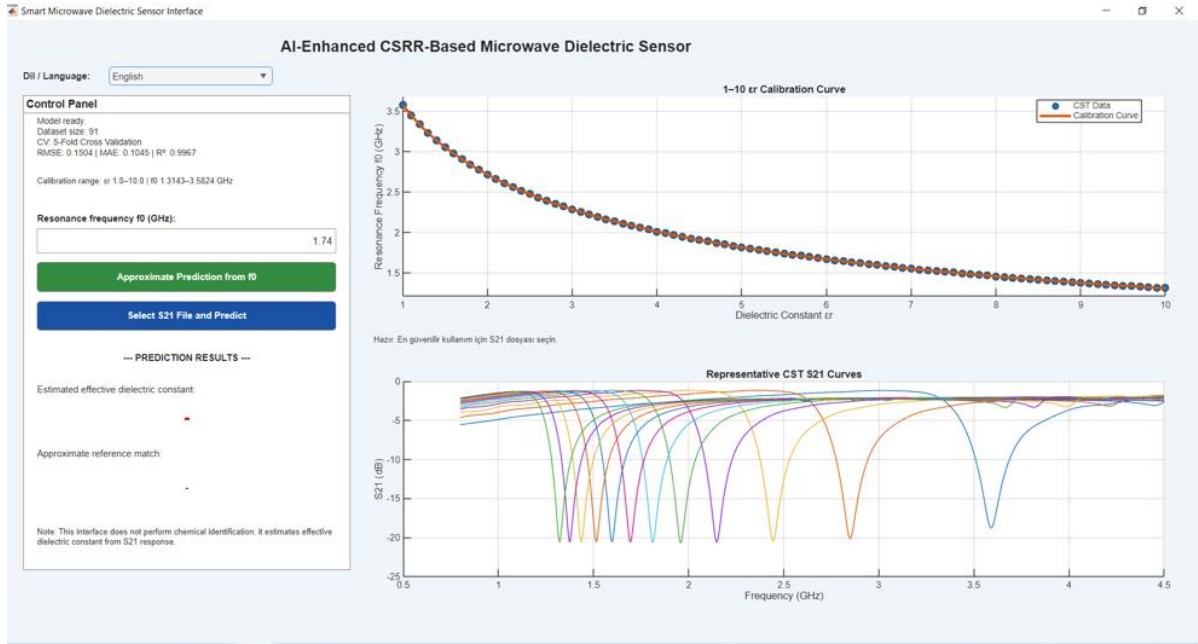


Figure 3.16. Prototype MATLAB interface showing the 1-10 epsilon calibration curve, representative CST S21 curves, and prediction panel.

3.10. Engineering Interpretation of the Results

The results satisfy the central physics-based expectation of the thesis. Increasing dielectric constant shifts the S21 resonance to lower frequency because the sample increases the effective capacitance of the resonant structure. The depth of the resonance remains strong across the sweep, which allows reliable extraction of the minimum point and bandwidth. The smooth f_0 -epsilon relation indicates that the same resonance branch was tracked over the full sweep rather than jumping between unrelated modes [1], [2].

The machine-learning results should be interpreted as numerical calibration performance. The model learns the inverse mapping from CST-generated S21 features to the input dielectric parameter. This is not equivalent to experimental accuracy in a fabricated device. Nevertheless, the result is significant because it demonstrates that the complete pipeline works: CST produces physically consistent S21 curves, MATLAB extracts meaningful descriptors, and the regression model accurately reconstructs the dielectric constant within the calibrated interval [4], [5], [13].

The project also shows why the corrected S21 workflow matters. If S11 had been exported while the thesis described S21, the sensing interpretation would be inconsistent. If the reference impedance had remained at a nonstandard value, comparison with RF measurement practice would be weaker. The final workflow corrects both issues by using S2,1 and 50 ohm normalization [1], [4], [13].

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3.11. Limitations of the Current Results

The current study is limited to simulation data. Fabrication tolerances, soldered connectors, VNA calibration uncertainty, substrate-property variation, temperature drift, and liquid-placement repeatability are not included. The model also estimates an effective real dielectric constant rather than a full complex permittivity spectrum. For this reason, the thesis does not claim definitive chemical identification. A future experimental version must use reference liquids, repeated measurements, uncertainty analysis, and possibly multi-frequency complex-permittivity estimation [1], [4].

4. CONCLUSIONS

This thesis developed a simulation-based AI-enhanced microwave dielectric characterization workflow using a two-port CSRR-based resonator, CST Studio Suite, MATLAB automation, S21 feature extraction, and Random Forest regression. The work is best interpreted as an end-to-end numerical calibration framework. It does not claim universal chemical identification; rather, it estimates the effective relative dielectric constant from the measured or simulated transmission response of a calibrated resonant structure [3], [5], [11], [14].

The electromagnetic interpretation is consistent with dielectric perturbation theory. Increasing relative permittivity increases the effective electric energy storage and capacitance of the resonator, which shifts the dominant S21 resonance toward lower frequency. The final 91-point sweep from $\epsilon_r = 1.0$ to 10.0 confirmed this expected behavior. The extracted calibration curve showed a smooth nonlinear inverse relationship between resonance frequency and dielectric constant [1], [4], [13].

The MATLAB-CST automation stage successfully generated 91 S21 response files with zero failed jobs. The S-parameters were normalized to a standard 50 ohm reference impedance, which is essential for RF measurement consistency. MATLAB extracted twelve physically meaningful descriptors from each S21 curve, including resonance frequency, minimum transmission, resonance depth, bandwidth, loaded Q, frequency shift, area, mean response, standard deviation, and local slopes [4], [5], [13].

The Random Forest regression model achieved strong numerical performance on the CST-generated dataset. The final 5-fold cross-validation results were RMSE = 0.150374 epsilon, MAE = 0.104530 epsilon, R-squared = 0.996723, and maximum absolute error = 0.541632 epsilon. These results demonstrate that the selected S21 features form an effective calibration vector for interpolation within the simulated dielectric range [3], [5], [11], [14].

The main limitation is that the present work is simulation-based. Physical fabrication, VNA measurement, liquid handling, temperature compensation, uncertainty analysis, and

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complex-permittivity estimation remain future steps. The next engineering stage should fabricate the resonator, measure known reference materials, compare measured and simulated responses, and retrain or correct the model using experimental data [1], [4], [13].

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APPENDIX A. MATLAB-CST S21 DATASET GENERATION CODE

The following MATLAB code block presents the corrected S21 dataset-generation logic in English and without explanatory comment lines.

```
function createCstS21Dataset()
clc; close all;
projectRoot = 'C:\Users\CEM\Desktop\Bitirme2';
cstFileName = 'dielektrik ile rezonatör.cst';
exportFolderName = 'S21_Exports';
epsilonParameter = 'eps_degeri';
epsilonValues = 1.0:0.1:10.0;
sParameterTreeItem = '1D Results\S-Parameters\S2,1';
cstPath = fullfile(projectRoot, cstFileName);
exportFolder = fullfile(projectRoot, exportFolderName);
if ~exist(cstPath, 'file')
    error('CST project file was not found: %s', cstPath);
end
if ~exist(exportFolder, 'dir')
    mkdir(exportFolder);
end
logPath = fullfile(exportFolder, 'CST_S21_export_log.txt');
diary(logPath);
diary on;
fprintf('\n===== \n');
fprintf('CST S21 DATASET GENERATION STARTED \n');
fprintf('Project root : %s \n', projectRoot);
fprintf('CST file : %s \n', cstPath);
fprintf('Export folder: %s \n', exportFolder);
fprintf('Sweep range : %.1f to %.1f with %.1f step \n', epsilonValues(1), epsilonValues(end),
epsilonValues(2)-epsilonValues(1));
fprintf('Result item : %s \n', sParameterTreeItem);
fprintf('===== \n \n');
cstApplication = [];
try
    cstApplication = actxserver('CSTStudio.application');
    invoke(cstApplication, 'OpenFile', cstPath);
    mws = invoke(cstApplication, 'Active3D');
    successfulFiles = 0;
    failedJobs = strings(0,1);
    for index = 1:numel(epsilonValues)
        epsilonNow = epsilonValues(index);
        exportPath = fullfile(exportFolder, sprintf('eps_%.2f.txt', epsilonNow));
        fprintf(['%03d/%03d] epsilon_r = %.2f \n', index, numel(epsilonValues), epsilonNow]);
        invoke(mws, 'StoreDoubleParameter', epsilonParameter, epsilonNow);
        invoke(mws, 'Rebuild');
        solver = invoke(mws, 'Solver');
        invoke(solver, 'Start');
        selected = selectCstTreeItem(mws, sParameterTreeItem, 30);
        if ~selected
            failedJobs(end+1,1) = sprintf('epsilon %.2f tree selection failed', epsilonNow);
            continue;
        end
        exporter = invoke(mws, 'ASCIIExport');
        invoke(exporter, 'Reset');
        invoke(exporter, 'FileName', exportPath);
        invoke(exporter, 'Execute');
        if exist(exportPath, 'file')
            successfulFiles = successfulFiles + 1;
            fprintf('Exported: %s \n', exportPath);
        else
            failedJobs(end+1,1) = sprintf('epsilon %.2f export file was not created', epsilonNow);
        end
    end
    invoke(mws, 'Save');
    fprintf('\n CST S21 DATASET GENERATION COMPLETED \n');
    fprintf('Successful file count: %d \n', successfulFiles);
    fprintf('Failed job count : %d \n', numel(failedJobs));
    if ~isempty(failedJobs)
```

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Yayın Tarihi: 28.04.2025

Revizyon Tarihi :

```

        disp(failedJobs);
    end
catch executionError
    fprintf('\nExecution failed: %s\n', executionError.message);
end
try
    if ~isempty(cstApplication)
        invoke(cstApplication, 'Quit');
        delete(cstApplication);
    end
catch
end
diary off;
end

function selected = selectCstTreeItem(mws, treeItem, maximumTry)
selected = false;
for trial = 1:maximumTry
    try
        invoke(mws, 'SelectTreeItem', treeItem);
        selected = true;
        return;
    catch
        pause(1.0);
    end
end
end
end

```

APPENDIX B. FEATURE EXTRACTION AND RANDOM FOREST TRAINING CODE

The following MATLAB code block reads the exported S21 curves, extracts physically interpretable features, trains the ensemble regression model, and writes the validation outputs.

```

function trainCsrrDielectricModel()
clc; close all;
projectRoot = 'C:\Users\CEM\Desktop\Bitirme2';
exportFolder = fullfile(projectRoot, 'S21_Exports');
outputFolder = fullfile(projectRoot, 'Nihai_Model_Ciktilari');
if ~exist(exportFolder, 'dir')
    exportFolder = uigetdir(projectRoot, 'Select the folder containing eps_*.txt files');
    if isequal(exportFolder, 0)
        error('No data folder was selected.');
```

```

    end
end
if ~exist(outputFolder, 'dir')
    mkdir(outputFolder);
end
files = dir(fullfile(exportFolder, 'eps_*.txt'));
if isempty(files)
    error('No eps_*.txt files were found in %s', exportFolder);
end
epsilonValues = zeros(numel(files),1);
for index = 1:numel(files)
    epsilonValues(index) = parseEpsilonValue(files(index).name);
end
[~, order] = sort(epsilonValues);
files = files(order);
rows = [];
rawCurves = struct('FileName', {}, 'Epsilon', {}, 'FrequencyGHz', {}, 'S21dB', {});
referenceF0 = NaN;
previousF0 = NaN;
for index = 1:numel(files)
    filePath = fullfile(files(index).folder, files(index).name);
    epsilonNow = parseEpsilonValue(files(index).name);
    data = readmatrix(filePath);

```

```

data = data(all(isfinite(data),2),:);
frequencyGHz = data(:,1);
s21dB = data(:,2);
feature = extractS21FeatureVector(frequencyGHz, s21dB, previousF0);
previousF0 = feature.F0_GHz;
if isnan(referenceF0)
    referenceF0 = feature.F0_GHz;
end
feature.DeltaF_GHz = feature.F0_GHz - referenceF0;
rows = [rows; {string(files(index).name), epsilonNow, feature.F0_GHz, feature.Smin_dB,
feature.Smax_dB, feature.Depth_dB, feature.Bandwidth3dB_GHz, feature.LoadedQ, feature.DeltaF_GHz,
feature.AreaBelowMax, feature.MeanS21_dB, feature.StdS21_dB, feature.LeftSlope_dBperGHz,
feature.RightSlope_dBperGHz}];
    rawCurves(end+1).FileName = files(index).name;
    rawCurves(end).Epsilon = epsilonNow;
    rawCurves(end).FrequencyGHz = frequencyGHz;
    rawCurves(end).S21dB = s21dB;
end
featureTable = cell2table(rows, 'VariableNames',
{'FileName', 'Epsilon', 'F0_GHz', 'Smin_dB', 'Smax_dB', 'Depth_dB', 'Bandwidth3dB_GHz', 'LoadedQ', 'DeltaF_GHz', 'A
reaBelowMax', 'MeanS21_dB', 'StdS21_dB', 'LeftSlope_dBperGHz', 'RightSlope_dBperGHz'});
featureNames =
{'F0_GHz', 'Smin_dB', 'Smax_dB', 'Depth_dB', 'Bandwidth3dB_GHz', 'LoadedQ', 'DeltaF_GHz', 'AreaBelowMax', 'MeanS21
_dB', 'StdS21_dB', 'LeftSlope_dBperGHz', 'RightSlope_dBperGHz'};
inputMatrix = featureTable{:, featureNames};
targetVector = featureTable.Epsilon;
rng(42);
model = fitrensemble(inputMatrix, targetVector, 'Method', 'Bag', 'NumLearningCycles', 300, 'Learners',
templateTree('MinLeafSize', 2));
cvModel = crossval(model, 'KFold', 5);
predicted = kfoldPredict(cvModel);
errorVector = predicted - targetVector;
metrics.RMSE = sqrt(mean(errorVector.^2));
metrics.MAE = mean(abs(errorVector));
metrics.R2 = 1 - sum(errorVector.^2) / sum((targetVector - mean(targetVector)).^2);
metrics.MaxAbsError = max(abs(errorVector));
metrics.CVType = '5-Fold Cross Validation';
save(fullfile(outputFolder, 'AkilliSensor_Model_Nihai.mat'), 'model', 'metrics', 'featureNames',
'featureTable');
writetable(featureTable, fullfile(outputFolder, 'ozellik_tablosu.csv'));
writeMetricsFile(fullfile(outputFolder, 'dogrulama_metrikleri.txt'), metrics, numel(targetVector),
numel(featureNames));
plotTrainingOutputs(featureTable, rawCurves, predicted, model, featureNames, outputFolder);
end

function epsilonValue = parseEpsilonValue(fileName)
token = regexp(fileName, 'eps_([0-9]+\.[0-9]*)', 'tokens', 'once');
if isempty(token)
    error('The epsilon value could not be parsed from %s', fileName);
end
epsilonValue = str2double(token{1});
end

function feature = extractS21FeatureVector(frequencyGHz, s21dB, previousF0)
searchMask = true(size(frequencyGHz));
if isfinite(previousF0)
    searchMask = abs(frequencyGHz - previousF0) <= 0.80;
end
if nnz(searchMask) < 5
    searchMask = true(size(frequencyGHz));
end
searchFrequency = frequencyGHz(searchMask);
searchS21 = s21dB(searchMask);
[minimumValue, localIndex] = min(searchS21);
resonanceFrequency = searchFrequency(localIndex);
maximumValue = max(s21dB);
depthValue = maximumValue - minimumValue;
halfPowerLevel = minimumValue + 3.0;
leftIndex = find(frequencyGHz < resonanceFrequency & s21dB <= halfPowerLevel, 1, 'first');
rightIndex = find(frequencyGHz > resonanceFrequency & s21dB <= halfPowerLevel, 1, 'last');
if isempty(leftIndex) || isempty(rightIndex) || rightIndex <= leftIndex

```

```

        bandwidth = NaN;
        loadedQ = NaN;
    else
        bandwidth = frequencyGHz(rightIndex) - frequencyGHz(leftIndex);
        loadedQ = resonanceFrequency / bandwidth;
    end
    areaValue = trapz(frequencyGHz, maximumValue - s21dB);
    meanValue = mean(s21dB);
    stdValue = std(s21dB);
    windowWidth = 0.06;
    leftMask = frequencyGHz >= resonanceFrequency - windowWidth & frequencyGHz < resonanceFrequency;
    rightMask = frequencyGHz > resonanceFrequency & frequencyGHz <= resonanceFrequency + windowWidth;
    leftSlope = fitLocalSlope(frequencyGHz(leftMask), s21dB(leftMask));
    rightSlope = fitLocalSlope(frequencyGHz(rightMask), s21dB(rightMask));
    feature.F0_GHz = resonanceFrequency;
    feature.Smin_dB = minimumValue;
    feature.Smax_dB = maximumValue;
    feature.Depth_dB = depthValue;
    feature.Bandwidth3dB_GHz = bandwidth;
    feature.LoadedQ = loadedQ;
    feature.DeltaF_GHz = NaN;
    feature.AreaBelowMax = areaValue;
    feature.MeanS21_dB = meanValue;
    feature.StdS21_dB = stdValue;
    feature.LeftSlope_dBperGHz = leftSlope;
    feature.RightSlope_dBperGHz = rightSlope;
end

function slopeValue = fitLocalSlope(x, y)
if numel(x) < 2
    slopeValue = NaN;
else
    coefficient = polyfit(x, y, 1);
    slopeValue = coefficient(1);
end
end

function writeMetricsFile(filePath, metrics, sampleCount, featureCount)
fid = fopen(filePath, 'w');
fprintf(fid, 'Data count      : %d\n', sampleCount);
fprintf(fid, 'Feature count    : %d\n', featureCount);
fprintf(fid, 'CV method       : %s\n', metrics.CVType);
fprintf(fid, 'RMSE            : %.8f epsilon\n', metrics.RMSE);
fprintf(fid, 'MAE             : %.8f epsilon\n', metrics.MAE);
fprintf(fid, 'R2              : %.8f\n', metrics.R2);
fprintf(fid, 'Maximum abs error: %.8f epsilon\n', metrics.MaxAbsError);
fclose(fid);
end

function plotTrainingOutputs(featureTable, rawCurves, predicted, model, featureNames, outputFolder)
figure('Color','w');
plot(featureTable.Epsilon, featureTable.F0_GHz, 'o-', 'LineWidth', 1.5);
grid on;
xlabel('Relative dielectric constant');
ylabel('Resonance frequency (GHz)');
title('Extracted f0-epsilon calibration curve');
saveas(gcf, fullfile(outputFolder, 'f0_epsilon_calibration.png'));
figure('Color','w');
scatter(featureTable.Epsilon, predicted, 45, 'filled');
hold on;
plot([min(featureTable.Epsilon) max(featureTable.Epsilon)], [min(featureTable.Epsilon)
max(featureTable.Epsilon)], 'k--', 'LineWidth', 1.2);
grid on;
xlabel('True epsilon');
ylabel('Predicted epsilon');
title('Cross-validated prediction performance');
saveas(gcf, fullfile(outputFolder, 'cv_predicted_vs_true.png'));
figure('Color','w');
scatter(featureTable.Epsilon, predicted - featureTable.Epsilon, 45, 'filled');
grid on;
xlabel('True epsilon');

```

```

ylabel('Prediction error');
title('Cross-validation error distribution');
saveas(gcf, fullfile(outputFolder, 'cv_error_distribution.png'));
end

```

APPENDIX C. PROTOTYPE SENSOR INTERFACE CODE

The following MATLAB code block gives the prototype interface layer used for visualization and demonstration. It does not change the scientific claim of the thesis; the model output remains an effective dielectric-constant estimate.

```

function smartMicrowaveSensorInterface()
clc; close all;
projectRoot = 'C:\Users\CEM\Desktop\Bitirme2';
outputFolder = fullfile(projectRoot, 'Nihai_Model_Ciktilari');
modelFile = fullfile(outputFolder, 'AkilliSensor_Model_Nihai.mat');
if ~exist(modelFile, 'file')
    error('The trained model file was not found: %s', modelFile);
end
loaded = load(modelFile, 'model', 'featureNames', 'featureTable', 'metrics');
model = loaded.model;
featureNames = loaded.featureNames;
featureTable = loaded.featureTable;
metrics = loaded.metrics;
figureHandle = uifigure('Name', 'Smart Microwave Dielectric Sensor Interface', 'Position', [80 80 1200
620], 'Color', [0.94 0.96 0.98]);
uilabel(figureHandle, 'Text', 'AI-Enhanced CSRR-Based Microwave Dielectric Sensor', 'Position', [360 575
520 28], 'FontSize', 18, 'FontWeight', 'bold', 'HorizontalAlignment', 'center');
controlPanel = uipanel(figureHandle, 'Title', 'Control Panel', 'Position', [20 40 310 515],
'BackgroundColor', [1 1 1], 'FontWeight', 'bold');
uilabel(controlPanel, 'Text', sprintf('Model ready\nDataset size: %d\nCV: %s\nRMSE: %.4f | MAE: %.4f | R2:
%.4f', height(featureTable), metrics.CVType, metrics.RMSE, metrics.MAE, metrics.R2), 'Position', [15 390
280 90], 'FontSize', 10);
uilabel(controlPanel, 'Text', 'Resonance frequency f0 (GHz):', 'Position', [15 330 260 24], 'FontWeight',
'bold');
frequencyField = uieditfield(controlPanel, 'numeric', 'Position', [15 300 275 28], 'Value',
featureTable.F0_GHz(round(height(featureTable)/2)));
predictButton = uibutton(controlPanel, 'Text', 'Approximate prediction from f0', 'Position', [15 250 275
34], 'BackgroundColor', [0.18 0.55 0.22], 'FontColor', [1 1 1], 'FontWeight', 'bold');
fileButton = uibutton(controlPanel, 'Text', 'Select S21 file and predict', 'Position', [15 205 275 34],
'BackgroundColor', [0.08 0.27 0.62], 'FontColor', [1 1 1], 'FontWeight', 'bold');
uilabel(controlPanel, 'Text', 'PREDICTION RESULTS', 'Position', [15 150 275 24], 'HorizontalAlignment',
'center', 'FontWeight', 'bold');
epsilonLabel = uilabel(controlPanel, 'Text', '-', 'Position', [15 95 275 45], 'FontSize', 18,
'FontWeight', 'bold', 'FontColor', [0.80 0.05 0.05], 'HorizontalAlignment', 'center');
matchLabel = uilabel(controlPanel, 'Text', 'Effective dielectric constant estimate only', 'Position', [15
35 275 45], 'FontSize', 10, 'HorizontalAlignment', 'center', 'WordWrap', 'on');
calibrationAxes = uiaxes(figureHandle, 'Position', [360 335 800 210]);
s21Axes = uiaxes(figureHandle, 'Position', [360 65 800 220]);
predictButton.ButtonPushedFcn = @(src,event) predictFromFrequency();
fileButton.ButtonPushedFcn = @(src,event) predictFromFile();
plotInterfaceBase();
function plotInterfaceBase()
    cla(calibrationAxes);
    scatter(calibrationAxes, featureTable.Epsilon, featureTable.F0_GHz, 22, 'filled');
    hold(calibrationAxes, 'on');
    plot(calibrationAxes, featureTable.Epsilon, smoothdata(featureTable.F0_GHz, 'movmean', 3),
'LineWidth', 1.5);
    grid(calibrationAxes, 'on');
    xlabel(calibrationAxes, 'Dielectric constant epsilon_r');
    ylabel(calibrationAxes, 'Resonance frequency f0 (GHz)');
    title(calibrationAxes, '1-10 epsilon calibration curve');
    cla(s21Axes);
    representative = unique(round(linspace(1, height(featureTable), min(12, height(featureTable)))));
    exportFolder = fullfile(projectRoot, 'S21_Exports');
    hold(s21Axes, 'on');
    for k = representative
        filePath = fullfile(exportFolder, char(featureTable.FileName(k)));
    end
end

```

```

        if exist(filePath, 'file')
            data = readmatrix(filePath);
            plot(s21Axes, data(:,1), data(:,2), 'LineWidth', 1.0);
        end
    end
    grid(s21Axes, 'on');
    xlabel(s21Axes, 'Frequency (GHz)');
    ylabel(s21Axes, 'S21 (dB)');
    title(s21Axes, 'Representative CST S21 curves');
end
function predictFromFrequency()
    f0 = frequencyField.Value;
    [~, nearestIndex] = min(abs(featureTable.F0_GHz - f0));
    inputVector = featureTable{nearestIndex, featureNames};
    inputVector(1) = f0;
    prediction = predict(model, inputVector);
    epsilonLabel.Text = sprintf('%.3f', prediction);
    matchLabel.Text = 'Approximate estimate based on the nearest calibrated feature vector';
end
function predictFromFile()
    [fileName, folderName] = uigetfile('*.txt', 'Select an exported S21 file');
    if isequal(fileName, 0)
        return;
    end
    data = readmatrix(fullfile(folderName, fileName));
    frequencyGHz = data(:,1);
    s21dB = data(:,2);
    feature = extractInterfaceFeatures(frequencyGHz, s21dB, featureTable.F0_GHz(1));
    inputVector = [feature.F0_GHz feature.Smin_dB feature.Smax_dB feature.Depth_dB
feature.Bandwidth3dB_GHz feature.LoadedQ feature.DeltaF_GHz feature.AreaBelowMax feature.MeanS21_dB
feature.StdS21_dB feature.LeftSlope_dBperGHz feature.RightSlope_dBperGHz];
    prediction = predict(model, inputVector);
    epsilonLabel.Text = sprintf('%.3f', prediction);
    matchLabel.Text = 'Prediction from selected S21 file';
    cla(s21Axes);
    plot(s21Axes, frequencyGHz, s21dB, 'LineWidth', 1.3);
    grid(s21Axes, 'on');
    xlabel(s21Axes, 'Frequency (GHz)');
    ylabel(s21Axes, 'S21 (dB)');
    title(s21Axes, ['Selected file: ', fileName]);
end
end
end

```